Omega 133 (2025) 103223

Contents lists available at ScienceDirect

Omega

journal homepage: www.elsevier.com/locate/omega

# Personalized recommendation, behavior-based pricing, or both? Examining privacy concerns from a cost perspective☆

Chi Zhou [a], Danyang Bai [a], Tieshan Li [b], Jing Yu [a,*]

[a] School of Management, Tianjin University of Technology, Tianjin 300384, China
[b] John Molson School of Business, Concordia University, Montreal, Quebec, H3G 1M8, Canada

ARTICLE INFO

*Keywords:*
Personalized recommendation
Behavior-based pricing
Privacy concerns
Privacy cost
Game theory

ABSTRACT

In the era of the big data, e-commerce increasingly adopts personalized recommendation and behavior-based pricing (BBP) strategies to enhance consumer experience, while also raising concerns about privacy. This study examines the impact of privacy costs on the effectiveness of those strategies using a two-period Hotelling model. The results indicate that retailers who combine personalized recommendation with BBP strategies can achieve higher prices and profits compared to those who do not employ these strategies, particularly when there are significant differences in privacy costs. Our study further reveals that relying solely on personalized recommendation without incorporating BBP may lead to decreases profit. Moreover, the accuracy of recommendations and variations in privacy costs significantly influence retailers' strategy choices, emphasizing the importance of these factors in gaining a competitive advantage. This research provides valuable insights for online retailers on how to effectively position themselves in the market while addressing consumer privacy concerns, offering a new perspective on the comprehensive impacts of personalized recommendation and BBP strategies in the business landscape.

## 1. Introduction

With the continuous evolution and integration of e-commerce platforms, consumer purchasing behavior is gradually shifting towards online channels. Unlike traditional brick-and-motor stores, online retailers find it easier to track consumers' personal information and purchase records [1,2]. Such feature makes it possible for online merchants to use big data algorithms, based on privacy information to provide personalized recommendation and implement differential pricing, i.e. behavior-based pricing (BBP). Behavior-Based Pricing (BBP) is a pricing strategy that involves analyzing and leveraging consumers' historical behavior and preferences to set prices [3]. This approach uses big data and advanced analytics to understand consumer behavior, thereby determining prices most likely to stimulate purchases [4]. Chen et al. [5] explored dynamic pricing strategies and analyzed how personalized pricing could be based on consumer strategic behavior. Unlike traditional cost-based or market competition-based pricing strategies, behavior-based pricing emphasizes personalization, aiming to maximize enterprise profits and consumer satisfaction through the analysis of consumer behavior and privacy data to provide differentiated prices [6].

Personalized recommendation technology is crucial for modern e-commerce, significantly enhancing user experience, optimizing sales efficiency, and increasing commercial value for business. Amazon uses personalized product recommendations based on individual purchase histories, extending to email notifications and mobile alerts, enhancing shopping experiences and satisfaction.[1] Netflix employs advanced algorithms to analyze user viewing histories and ratings, delivering personalized movie and TV show recommendations, boosting user engagement and loyalty.[2] Sephora's "Beauty Insider" program collects beauty preferences, skincare details, and purchase histories to tailor personalized

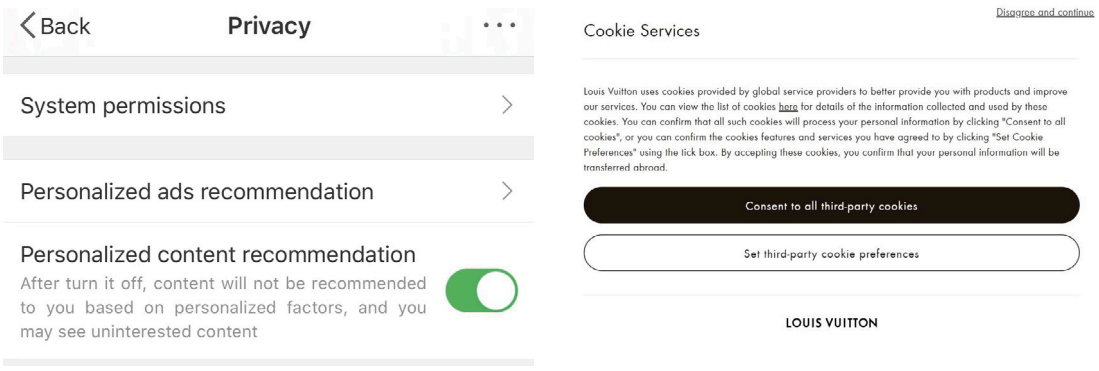
☆ Area: Supply Chain Management This manuscript was processed by Associate Editor Bhaskaran.

* Corresponding author.
*E-mail addresses:* czhou@tju.edu.cn (C. Zhou), dyangbai@163.com (D. Bai), tieshan.li@concordia.ca (T. Li), yujing@tjut.edu.cn (J. Yu).

[1] For instance, a consumer who searches or purchases a sport shirt from www.amazon.com gets personalized recommendation and prices of shirts with matching similar tastes.

[2] After logging in, users see a "Recommended for You" section that includes personalized content suggestions based on their viewing history and preferences. (See https://www.netflix.com).

https://doi.org/10.1016/j.omega.2024.103223
Received 4 April 2024; Accepted 26 October 2024
Available online 11 December 2024
0305-0483/© 2024 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC license (http://creativecommons.org/licenses/by-nc/4.0/).




(a) Personalized recommendation option. (b) LV with BBP set privacy concerns option.

**Fig. 1.** Consumers have the autonomy to choose how to protect their privacy.

product recommendations, such as skincare products for specific skin types.[3] These examples illustrate how personalized recommendations improve shopping experiences, satisfaction, sales, customer loyalty, and marketing effectiveness for companies. Purchasing history often includes sensitive information such as consumers' payment details and buying behaviors, which can be more personal than search history [7,8]. Consumers tend to manage their information through privacy options, thereby affecting how retailers tailor their recommendations and pricing strategies.

A survey shows that 78.5 percent of online retailers collect consumer browsing history and search data for advertising recommendation and third-party sales [9]. Target, the eighth largest online retailer in the United States, collects consumer privacy data during the purchasing process, including personal identity, search preferences, device identification, and geographic location. This enables Target to practice personalized product recommendation to consumers [10]. Similarly, the luxury brand Louis Vuitton uses collected consumer identity information to recommend premium-priced products and services to frequent patrons of upscale venues, and sets varied prices for consumers based on their purchasing levels [11]. However, while consumers benefit from improved and tailored recommendation services, they are concerned about privacy breaches and possible price discrimination. Consequently, retailers employing personalized approaches must take proactive measures to and alleviate address consumer privacy concerns [12].

In fact, if online retailers can access consumer information, personalized recommendation and BBP strategies become viable options [13]. To address privacy concerns, online retailers often offer consumers the option to opt in or out of personalized recommendation through their privacy settings interface, as shown in Fig. 1(a). Consumers who wish to receive personalized recommendations, can activate this option, while those who prefer not to can opt out. Additionally, consumers can also choose the degree of privacy for information sharing through the cookie settings provided by online retailers, depicted in Fig. 1(b). If consumers who are comfortable sharing all their privacy information can opt for the "Agree to All" option. Alternatively, they can choose to disclose only certain information based on their preferences. After consumers make their choices, online retailers can categorize them into different groups based on the information collected during the purchasing process. These groups typically include new and repeat consumers, allowing online retailers to implement personalized recommendation and BBP strategies [3].

An increasing number of consumers are aware of the privacy costs associated with data breaches in online shopping. Privacy costs refer to the costs consumers face when they choose to share personal information during the purchase of products or services [14]. These costs vary among different retailers, leading consumers to carefully consider the differences in privacy costs when making purchases to protect their privacy. The General Data Protection Regulation (GDPR), effective May 25, 2018, is an EU framework that imposes strict privacy requirements on online retailers for data collection, storage, and processing to ensure security and confidentiality [15]. Retailers must comply with GDPR to avoid legal risks in personalized recommendations and behavioral targeting. Since GDPR's enactment, data breaches have notably decreased, indicating improved personal data protection. Consumers who leverage privacy concerns can prevent price discrimination and avoid third-party data sales [16,17]. However, they may miss out on accurate personalized recommendations, leading to a reduced service experience.

Although personalized recommendation and behavior-based price discrimination (BBP) present viable options, determining the optimal strategy for online retailers remains challenging. Therefore, this research explores personalized recommendation and BBP within the context of privacy concerns. Given that consumers have control over their privacy preferences, we propose the following research questions: (a) What criteria do retailers employ when choosing personalized recommendation and BBP strategies? (b) How do prices and profits of online retailers change over two periods under varying strategies? (c) To what extent do factors such as the recommendation service level and privacy cost impact retailers' strategies?

To answer these questions, we propose a two-period duopoly game model where two online retailers sell symmetric products and compete in the market through personalized recommendation and BBP. In the market, personalized recommendation and BBP strategies are used to increase sales and profits while respecting consumer privacy. Amazon tailors product recommendations to based on customer's behavior and adjusts prices according to individual preferences. For privacy-conscious, high-value buyers, it may present high-priced items, reflecting an assumption of a greater willingness to pay. For price-sensitive users, Amazon provides personalized recommendation accordingly. This strategy optimizes revenue and customer satisfaction, reflecting the effectiveness of Amazon's data-driven methods.[4] When consumers purchase products or services from

[3] For instance, users may receive personalized product recommendations on their account page or via email. Based on consumers' purchase history and preferences, Sephora may offer customized discounts and special promotions. (See https://www.sephora.com).

[4] On the product and cart pages, Amazon displays recommended products. After consumers purchase items based on these recommendations, Amazon may offer customized discounts and promotions based on their shopping behavior, which constitutes a form of price discrimination to some extent. (See https://www.amazon.com).

retailers, they do not always follow personalized recommendations. Considering consumers' increased concern for privacy, retailers offer them personalized choices, as illustrated by Herms and Louis Vuitton in Fig. 1. If consumers opt to protect their privacy, retailers are unable to use personalized recommendations and BBP strategies. However, once consumers opt into personalized options, two scenarios emerge: retailers either provide personalized recommendations only or engage in personalized pricing alongside personalized recommendation [18]. To include all possible scenarios, we study three models, namely, Model NN (without personalized recommendation and BBP), Model BN (with personalized recommendation and without BBP), and Model BB (with both personalized recommendation and BBP). This paper aims to examining the impact of personalized recommendation and BBP strategies while taking consumers privacy concerns into account.

The main contributions of this paper are as follows. First, we identify the optimal personalized recommendation and BBP strategies for online retailers. In general, personalized recommendation and BBP are beneficial to online retailers. After comparing the equilibrium results across various scenarios, we conclude that Models NN and BB yield the most effective strategies. When considering privacy concerns, if consumers protect their privacy, retailers cannot use personalized recommendation and BBP. Nonetheless, in this case, retailers can still obtain profits. Conversely, when consumers opt to share their privacy, retailers can choose between personalized recommendation and BBP. If retailers only provide personalized recommendation, the unified pricing decision may increase retailers' total profits to some extent, but not as effectively as the other two models. However, if retailers provide both personalized recommendation and BBP to consumers, they gain a competitive edge in the market across both periods. Therefore, the optimal strategy of online retailers and their competitors is either to adopt both personalized recommendation and BBP, or neither of them.

Subsequently, we investigate the dynamic evolution of online retailers' optimal prices and profits across different strategies. Employing mathematical models, we meticulously compare retailers' equilibrium outcomes over two distinct periods. Our analysis consistently reveals a strategic pricing pattern: retailers strategically set higher prices during the initial period, followed by reductions in subsequent periods to stimulate consumer demand. Notably, our research unveils that online retailers relying solely on personalized recommendation tend to adopt a low-pricing strategy. This phenomenon can be attributed to the limited effectiveness of personalized recommendation in generating substantial market demand. To offset this limitation, retailers choose price reductions as an incentive to attract consumers. In contrast, online retailers who combine personalized recommendation with BBP experience a gradual increase in both prices and profits. This notable performance underscores the synergistic benefits of integrating these two strategies, highlighting their complementary strengths in driving business performance.

Furthermore, our investigation unveils the significant impact of the recommendation service level and privacy cost difference on online retailers' strategies. Consumers generally prioritize optimizing their shopping experiences, thereby being willing to accept higher privacy costs associated with enhancing those experiences. This preference leads retailers to adjust their product and service prices accordingly when improving recommendation service levels. Specifically, as recommendation services become more personalized and accurate, consumers perceive increased value, thus becoming willing to pay higher prices. This causal relationship indicates that enhancing recommendation service levels not only enhances consumers' shopping experiences but also enables retailers to reflect the value of these improvements through price adjustments.

However, the escalating recommendation cost prompts a growing number of consumers to compromise their privacy concerns in favor of shopping convenience. This shift in consumer behavior provides an opportunity for retailers to leverage privacy data, thereby leading to increased profits. Considering privacy concerns, the expanding gap in privacy cost intensifies market competition. In this context, personalized recommendation and BBP strategies emerge as effective tools for retailers to secure competitive advantages and capture greater market share. These findings underscore the critical importance for retailers to carefully select strategies that foster mutually beneficial outcomes for both themselves and consumers. Therefore, both the recommendation service level and privacy cost difference generally favor online retailers.

The remainder of this paper is structured as follows. Section 2 presents a review of the relevant literature. In Section 3, we elaborate on the fundamental research model and parameter configurations. In Section 4, we analyze the distinct equilibrium outcomes under the three models. Section 5 compares the three models and provides intuitive explanations. Section 6 investigates the impact of recommendation service level and privacy cost difference on the equilibrium outcomes of the online retailer. We provide the managerial implication for the results in Section 7. In Section 8, we have added two extensions to make the study more comprehensive. Finally, Section 9 concludes this paper and suggests directions for future research. All omitted proofs are included in Appendix.

## 2. Literature review

This study primarily addresses three categories of relevant literature.

### 2.1. Personalized recommendation

Firstly, our research is closely related to the literature on personalized recommendation. Personalized recommendation involves utilizing big data to conduct tailored product promotion, thereby attracting more consumers to purchase products or experience services online. Choi et al. [19] has developed a recommender system to facilitate two-way information exchange between consumers and relevant online retailers, thus enabling more dynamic personalized recommendation. With the emergence of new interest factors such as social networks, a unified personalized recommendation model based on probabilistic matrix factorization is proposed to incorporate social factors such as personal interests [20]. Zhou and Han [21] investigate ranking-oriented recommendation algorithms, with a specific focus on how consumer preferences influence the diversity of personalized recommendation. Cheah et al. [22] primarily examine how online retailers select personalized recommendation strategies and provide recommendation to consumers across two periods, aiming to maintain strong consumer relationships and enhance consumer loyalty [23]. When alternative products emerge and the platform provides online recommendation services, retailers investigate the implications for online retailers considering the introduction of personalized recommendation marketing channels [24]. They explore how these channels could enhance consumer engagement and sales. Businesses leverage advanced algorithms to effectively exploit consumer privacy preferences with the aim of optimizing the resource allocation process [15]. Their work aligns target users more accurately with the products or services offered by online retailers. This approach not only improves the efficiency of personalized recommendations but also enhances the overall shopping experience by better matching consumer interests with available options. Guo et al. [25] further build on this by examining the impact of these refined strategies on the effectiveness of personalized recommendations, providing insights into optimizing marketing practices in the digital age.

Prior scholars have extensively examined personalized recommendation systems and their impact on retailers' competitive convergence, price discrimination, and conversion of purchase intention in the existing literature. Their major focus lies in analyzing the relationship between

personalized recommendation diversity and consumer behavior [26]. In contrast, our research places emphasis on consumer initiatives regarding privacy sharing and highlights retailers' strategies in providing varying degrees of product recommendation to new and existing consumers. This research model demonstrates a higher degree of innovation compared to most alternatives. Our findings suggest that online retailers should be attentive to consumers' privacy concerns and strategically select personalized recommendation services accordingly.

### 2.2. Behavior-based price discrimination

Concerning the behavior of new and existing consumers, retailers often practice price discrimination by tailoring pricing based on private information. Some scholars also explore the impact of factors such as durable products, long-term contracts, and age identification on BBP. For example, Villas-Boas [27] investigates how retailers capture private information in consumers' purchase history and choose to use BBP to personalize pricing for both new and existing consumers. Looking at it from another perspective, online retailers switch to a market composed of forward-looking consumers to sell to paying consumers [5]. Villas-Boas [28] demonstrates how retailers can utilize BBP to reward repeat consumers with lower prices than new consumers. Cui et al. [29] study a price discrimination model of trade promotion, revealing an incentive for price discrimination between dominant retailers and smaller independent retailers. Zhang [30] explores the possibility of retailers offering new consumers higher quality than existing consumers. If consumers face low switching cost, exhibit diverse needs and evolving preferences, or discount future purchases, price discrimination can be allowed to encourage market entry and enhance welfare [31]. In markets without vertical-differentiated markets, retailers that track consumers' personal information often use behavior-based pricing (BBP), which discriminates against consumers based on preferences revealed in their private information [32]. Li [33] studies various sources of manufacturers and retailers when choosing to use BBP, how to determine the optimal wholesale and retail prices, and implementation of the industry's total profits, consumer surplus and social benefits. Concurrently, Choe et al. [4] explore a model of dynamic competition between two retailers, focusing on their acquisition and utilization of consumer information in the first period.

In a competitive market, competitive intensity significantly influences the decisions of retailers and consumers. High competitive markets offer consumers more choices, prompting them to focus on price and quality, thereby driving retailers to offer competitive prices and improve product quality [34]. Conversely, in markets with lower competitive intensity, consumer choices may be more limited, potentially relying more on brand loyalty. This may compel retailers to adopt aggressive strategies such as price reductions and enhanced services in high-competition environments, whereas in low-competition environments, they may maintain higher prices and profits but could reduce innovation [35]. Due to the interdependent nature of their pricing strategies, lack of strategic coordination may lead both parties to adopt suboptimal strategies, resulting in a prisoner's dilemma [36]. When two retailers implement BBP, they may compete for market share by either lowering prices or enhancing the precision of their recommendation systems, triggering a price war that ultimately reduces profits. If companies have better insights into consumer purchase histories and preferences than consumers have about pricing strategies, they may exploit short-term gains from asymmetric information [37]. However, this could potentially damage consumer trust and market fairness in the long run. From the perspective of emerging subscription service models, Zhou et al. [38] examine the ability of retailers to continuously improve the quality of their online services based on consumers' private information. They find that when the differences in quality improvement between two retailers are minimal, multiple equilibria emerge, with both retailers benefitting from the BBP model.

Our study integrates the BBP model with personalized recommendation strategies in a horizontally differentiated market and introduces privacy cost difference to elucidate the final equilibrium outcomes. We argue that consumers possess the capability to manage their privacy strategically, such as through the selective disclosure of consumer data and considerations of privacy concerns, which have not been examined in previous research by excluding factors like contract duration, pricing agreements, or unconventional channels. As the disparity in privacy costs increases, personalized pricing does not necessarily provoke intense price competition, thus enriching the existing research on BBP. Concurrently, our study examines the influence of recommendation service level on the selection of personalized recommendation and BBP strategies, offering valuable insights for online retailers. The personalized pricing we study involves retailers like Anta or Xtep[5] setting prices based on specific attributes or behavioral histories of individual consumers, aimed at maximizing each consumer's willingness to pay. Dynamic pricing, on the other hand, involves retailers adjusting prices based on market conditions, competitive dynamics, or supply–demand relationships to reflect market changes and optimize sales strategies [37]. Furthermore, we investigate the impact of BBP on prices and profits across two periods, presenting a novel perspective on the feasibility of combining personalized recommendation and price discrimination.

### 2.3. The privacy concerns

The third category of literature focuses on privacy issues, an area that has increasingly become a focal point in academia as consumer awareness of personal information protection grows. In this realm of research, Acquisti and Varian [39] provide a comprehensive overview of the economic value of privacy through their work, establishing a solid foundation for subsequent privacy-related studies. Their research not only reveals the importance of privacy in economic activities but also offers a theoretical framework for understanding how consumers balance privacy concerns with convenience. Following Acquisti and Varian [39]'s work, Goldfarb and Tucker [40] extensively explore advertisers' capabilities in using online consumer data and analyze the impact of such data-driven advertising on market competition. Their study emphasizes how data-driven advertising alters market structures and presents new challenges for consumer privacy protection. Conitzer et al. [41] further expand research in this field, focusing on pricing strategies adopted by online retailers when faced with privacy-sensitive consumers. Their work reveals how privacy considerations influence consumer purchase decisions and how retailers adjust pricing strategies to attract privacy-sensitive consumers. Casadesus-Masanell and Hervas-Drane [42] examine the impact of consumer privacy issues on market competition in a bilateral monopoly model, where the extent of privacy disclosure is treated as an endogenous choice. Their research provides new insights into strategic choices for firms in balancing consumer privacy protection with data utilization to enhance market competitiveness. Shy and Stenbacka [43] propose a switching cost model by considering varying degrees of privacy issues and their effects on industry profits, consumer surplus, and overall welfare. Their work demonstrates that privacy issues not only affect consumers and businesses but also have profound implications for societal welfare.

[5] Anta and Xtep are well-known sportswear brands in China. They may employ different pricing strategies on their official websites or e-commerce platforms, including promotions and discounts. These strategies can, to some extent, be considered a form of price discrimination.

Montes et al. [1] empirically investigate the relationship between consumer acceptance of personalized advertising and their sense of control and benefits concerning privacy issues. Their findings provide empirical foundations for understanding how consumers protect their privacy rights while enjoying personalized services. Ichihashi [44] analyzes the impact of consumer privacy issues on benefits and prices, particularly in cases where consumers disclose personal information in exchange for personalized recommendations. This study offers new insights into the motivations and consequences of privacy disclosures. Finally, Chen and Duan [45] propose new research directions, contributing new theoretical and empirical analyses to the study of privacy issues. Together, these studies form the foundation of our understanding of privacy issues and provide rich resources and multidimensional perspectives for future research. As technology advances and consumer behaviors evolve, privacy issues continue to be a hotspot of concern in both academia and practice.

There are certain limitations in prior research on privacy concerns. Studies indicate that sellers often incentivize information sharing by promising not to utilize the information for pricing purposes. These studies also reveal that consumers can obtain greater utility by concealing certain personal information. Additionally, weaker privacy concerns tend to benefit online retailers more than either strong or nonexistent privacy concerns. Our research takes the perspective that consumers can safeguard their privacy based on the options provided by retailers. We use a model featuring relatively rational consumers who continuously assess the benefits of privacy concerns. This study is crucial as it explores how privacy concerns influence the personalized recommendation and pricing strategies of online retailers, highlighting the significant relationship between consumers' privacy concerns and these retail strategies.

Lastly, the research most relevant to our study is Li [46], which studies the impact of price discrimination through the Hotelling model in the context that retailers use new technologies to identify new and existing consumers based on consumers' purchase information. Similarly, we find that implementing BBP can increase industry profits, and consumers make strategic choices regarding their privacy concerns. In our model, two symmetric online retailers gather consumer information from their choices and employ personalized recommendations and BBP strategies influenced by these privacy concerns. Thus, we develop the following model to further investigate these dynamics.

## 3. The model

We propose a two-period model of duopoly competition, where two symmetric online retailers, A and B, compete in the product market. We assume that the utility consumers derive from purchasing products from retailers (represented as $v$) is sufficiently high, indicating that the market can be fully covered. In general, online retailers can obtain certain privacy information before consumers place orders, relying on various user portals such as website registration, product and service logs, and so on [47]. As mentioned in the introduction, our research addresses privacy concerns and further defines them as a privacy option. In fact, consumers have the right to decide whether to protect their privacy, i.e., they can choose to enable or disable that option. It is important to note that when the privacy option is enabled, the retailer offers personalized recommendation or takes the BBP strategy based on that. Therefore, our model does not consider scenarios where retailers implement BBP without incorporating personalized recommendations.

Our model employs the Hotelling model, wherein consumers are uniformly distributed along the Hotelling line between 0 and 1. These consumers exhibit heterogeneous preferences for the product, which are reflected in their locations on the Hotelling line. Without loss of generationality, we assume that the marginal cost of the product is zero and that there exists a unit mass of consumers who engage in purchase over two periods within a duopoly market [48]. Consumers show a stronger preference for products with a higher recommendation service level and a lower price, without having any inherent preferences prior to purchasing the product. Consumers hold rational expectations regarding the retailers' strategies across these two periods, implying that their choices in the first period are influenced by the second period. Additionally, retailers observe varying purchasing behaviors and demands based on consumers' voluntarily shared privacy information and purchasing data [49].

From the perspective of consumers, we take into account their privacy concerns. When consumers choose to protect privacy, they do not incur the cost of privacy. According to Li [33], retailers have the option to forgo both personalized recommendation and BBP. The utility of consumers located at $x$ to purchase from retailer A is represented as $v - x - p_{aj}$, while the utility of purchasing from retailer B is $v - (1 - x) - p_{bj}$, where $p_{ij}$ denotes the price set by the retailer $i$ in period $j$ ($i \in \{a, b\}, j \in \{1, 2\}$). When consumers activate the privacy switch to access personalized services, they suffer the loss of privacy, defined here as the cost of disclosing privacy information. We denote this cost as $c_i$. To simplify the calculation, we use $c_\Delta$ to represent the difference in privacy cost between online retailers A and B, i.e., $c_\Delta$ is $c_b - c_a$.

Retailers are able to adopt personalized recommendation and BBP strategies when consumers activate the privacy switch. Online retailers can enhance their profits by increasing the recommendation service level. We assume that the retailer's cost of providing recommendation services to consumers is related to the recommendation service level. We use $c$ to denote the retailer's unit cost of making recommendation, and $r_i$ to denote the retailer's recommendation service level for the consumer. Thus, the recommendation cost for the retailer is $cr_i$. In fact, when online retailers adopt personalized recommendation, consumers exhibit varying sensitivities to recommendation. We use $\beta$ to represent the sensitivity of personalized recommendation. Thus, the effect of personalized recommendation service level on consumers is $\beta r_i$. In the second period, compared to existing consumers, new consumers are more susceptible to the influence of recommendation service level. This is primarily because they have less understanding of market choices and products, and they require more personalized advice and information to aid their decision-making [26]. This enhances the impact and effectiveness of recommendation systems in attracting and influencing new consumers. Therefore, new consumers are more likely to be affected by the recommendation service level. We set the personalized recommendation sensitivity for new consumers as $\beta = 1$. However, existing consumers are less influenced by the recommendation service level, with the personalized recommendation sensitivity range of $0 < \beta < 1$.

The sequence of the game is as follows. The first period comprises three stages. Initially, the consumer makes a decision on privacy concerns, opting to either enable or disable the privacy option. Next, the online retailer sets the price and fundamental recommendation service level. Finally, the consumer selects the retailer for product purchase based on the retailer's strategy. The second period also consists of three stages. Firstly, the online retailer decides whether to adopt personalized recommendation and BBP. In the second stage, if the retailer solely adopts personalized recommendation, they offer a uniform price to all consumers and determine recommendation service level. If the retailer additionally adopts BBP, $p_{io}$ represents the price charged to existing consumers, while $p_{in}$ denotes the price for new consumers. In the third stage, consumers decide whether to continue purchasing from the original retailer, or switch to the other retailer being new customers.

We summarize the parameter notations in Table 1.

**Table 1**
List of notations and descriptions. ($i \in \{a, b\}, j \in \{1, 2\}$)

| Notation | Description |
|---|---|
| $v$ | Base value of the product and service |
| $x$ | Location of consumers on the Hotelling line |
| $r_i$ | Recommendation service level |
| $c$ | Recommendation cost per unit |
| $c_i$ | Privacy cost |
| $\beta$ | Personalized recommendation sensitivity |
| $p_{ij}$ | Online retailer $i$'s price in period $j$ |
| $p_{io}$ | Online retailer $i$'s price for existing consumers with BBP |
| $p_{in}$ | Online retailer $i$'s price for new consumers with BBP |
| $\pi_{ij}$ | Profit made by the online retailer $i$ in period $j$ |

**Table 2**
Summary of Two-Period Equilibrium Prices and Model Construction.

| Model | Period 1 | Period 2 |
|---|---|---|
| $NN$ | $p_{a1}^{NN*} = p_{b1}^{NN*} = 1$ | $p_{a2}^{NN*} = p_{b2}^{NN*} = 1$ |
| $BN$ | $p_{a1}^{BN*} = \frac{14+6c_\Delta+3(1+\beta)r_a-3(1+\beta)r_b}{21}$<br>$p_{b1}^{BN*} = \frac{14-6c_\Delta-3(1+\beta)r_a+3(1+\beta)r_b}{21}$ | $p_{a2}^{BN*} = \frac{7+4c_\Delta+2(1+\beta)r_a-2(1+\beta)r_b}{14}$<br>$p_{b2}^{BN*} = \frac{7-4c_\Delta-2(1+\beta)r_a+2(1+\beta)r_b}{14}$ |
| $BB$ | $p_{a1}^{BB*} = \frac{32+6c_\Delta+(11-5\beta)r_a+(5-11\beta)r_b}{24}$<br>$p_{b1}^{BB*} = \frac{32-6c_\Delta+(5-11\beta)r_a+(11-5\beta)r_b}{24}$ | $p_{ao}^{BB*} = \frac{32+18c_\Delta+(1+17\beta)r_a-(17+\beta)r_b}{48}$<br>$p_{an}^{BB*} = \frac{8+6c_\Delta-(-7+\beta)r_a+r_b-7\beta r_b}{24}$<br>$p_{bo}^{BB*} = \frac{32-18c_\Delta-(17+\beta)r_a+r_b+17\beta r_b}{48}$<br>$p_{bn}^{BB*} = \frac{8-6c_\Delta+(1-7\beta)r_a-(-7+\beta)r_b}{24}$ |

## 4. Equilibrium results under various models

Based on the above explanation in Section 3, we establish three models. When consumers deactivate the privacy option, retailers are unable to utilize personalized recommendation and BBP (Model NN). On the contrary, retailers can access consumers' privacy, adopting personalized recommendation only (Model BN) or providing consumers with both personalized recommendation and BBP (Model BB). In this section, we analyze the equilibrium outcomes of these three scenarios.

### 4.1. Model NN: Without personalized recommendation and BBP

Before formally discussing the patterns of online retailers choosing personalized recommendation and BBP, we first examine the benchmark model. In this model, retailers cannot choose personalized recommendation and BBP because consumers choose to protect privacy. The two-period game without personalized recommendation and BBP is similar to static competition when consumers choose to protect their private data. By computing the equality of consumer utility, we obtain the location of the indifference point for online retailers A and B as $x_n$.

$$v - p_{a2} - x_n = v - p_{b2} - (1 - x_n), \tag{1}$$

$$x_n = \frac{1 + p_{b2} - p_{a2}}{2}. \tag{2}$$

The profit of retailer A is $\pi_{a2} = p_{a2}x_n = \frac{p_{a2}(1+p_{b2}-p_{a2})}{2}$. Similarly, the profit of retailer B is $\pi_{b2} = p_{b2}(1 - x_n) = \frac{p_{b2}(1+p_{a2}-p_{b2})}{2}$. We find that the equilibrium results for both periods in Model NN are very similar, so we do not repeat them below. Using the first order derivatives, we derive the equilibrium results for retailers A and B in both periods in Table 2.

### 4.2. Model BN: With personalized recommendation and without BBP

Now we consider a model where online retailers have access to data that gives them the option to provide personalized recommendation to consumers who opt to disclose their private information, but without using a BBP strategy. We employ backward deduction in this analysis.

**The Second Period**. We denote the marginal consumer's choice between online retailers A and B in the second period by the indifference point $x_1$. We have

$$v - p_{a2} - x_1 + \beta r_a - c_a = v - p_{b2} - (1 - x_1) + r_b - c_b, \tag{3}$$

$$x_1 = \frac{(\beta r_a - r_b) + p_{b2} - p_{a2} + 1 + c_\Delta}{2}. \tag{4}$$

The left side of Eq. (3) describes the utility of consumer repurchasing the same product after purchasing it from retailer A in the first period, and the right side represents the consumer utility achieved by purchasing from retailer B for the first time. Furthermore, a returning consumer of retailer A receives a differentiated recommendation service level $\beta r_a$ and uniform pricing $p_{a2}$ set by the online retailer A. Conversely, a consumer who makes a purchase from retailer B can be regarded as a new consumer and receives personalized recommendation, where the recommendation service level is $r_b$ and the price is $p_{b2}$.

We denote the point by $x_2$ whether the marginal consumer is indifferent between switching to online retailer A or B. This yields

$$v - p_{a2} - x_2 + r_a - c_a = v - p_{b2} - (1 - x_2) + \beta r_b - c_b, \tag{5}$$

$$x_2 = \frac{(r_a - \beta r_b) + p_{b2} - p_{a2} + 1 + c_\Delta}{2}. \tag{6}$$

The left and right sides of Eq. (5) are the utility received by the consumer making a purchase from retailers A and B, respectively. If we consider the recommendation cost to be paid in this model, we get the profit formulas of online retailers as

$$\pi_{a2} = p_{a2}x_1 + p_{a2}(x_2 - x_n) - cr_a, \tag{7}$$

$$\pi_{b2} = p_{b2}(1 - x_2) + p_{b2}(x_n - x_1) - cr_b. \tag{8}$$

**The First Period**. Consumers make their first purchase in Period 1, expecting a better product experience. Since the consumer is forward looking and is able to anticipate the pricing in the second period based on his/her decision in the first period, the expected price he/she chooses is denoted by $p_{a2}^{e*}$ or $p_{b2}^{e*}$. Let $x_n$ denote the indifference point in the first period, located on the Hotelling line. The left and right sides of Eq. (9) represent the sum of the utilities of retailers A and B in two periods, respectively. We have

$$v - p_{a1} - x_n + r_a - c_a + [v - p_{b2}^{e*} - (1 - x_n) + r_b - c_b] = v - p_{b1} - (1 - x_n) + r_b - c_b + [v - p_{a2}^{e*} - x_n + r_a - c_a] \tag{9}$$

Substituting $p_{a2}(x_n)$ and $p_{b2}(x_n)$ into the consumer's utility equation, we get

$$x_n = \frac{1 + 2c_\Delta - 3p_{a1} + 3p_{b1} + (1 + \beta)r_a - (1 + \beta)r_b}{2}. \tag{10}$$

Since retailers' decisions are influenced by the second period, they need to add the expected profit in the second period to their first period profit equation. Then, we obtain their profit functions as

$$\pi_{a1} = p_{a1}x_n + \pi_{a2}^{e*}, \tag{11}$$

$$\pi_{b1} = p_{b1}(1 - x_n) + \pi_{b2}^{e*}. \tag{12}$$

Based on the research process outlined above, we consolidate the equations and conduct the corresponding calculations to derive Table 2. In Corollary 1, we compare the equilibrium prices and profits in two periods in Model BN.

**Corollary 1.** *In Model BN, retailers adopt personalized recommendation only. We have*

*(i) The prices in period 2 are lower than those in period 1.*
*(ii) If $\tilde{c}_1(\tilde{c}_3) < c_\Delta < \tilde{c}_2(\tilde{c}_4)$, the retailer A(B)'s profit in period 2 is higher than that in period 1. Otherwise, the profit in period 2 is lower than that in period 1.*

Based on Corollary 1, we analyze Model BN across two periods, noting the retailer's use of personalized recommendation. The retailer's second-period prices are lower regardless of recommendation service level due to consumers' preferences shaped by first-period feedback, favoring lower prices. The retailer's second-period profit rises with the privacy cost difference $c_\Delta$ until it peaks at an intermediate level. Beyond this point, profit declines as increasing consumer sensitivity to privacy costs and product prices raise the recommendation costs for retailers, potentially causing profits to drop below the levels in the first-period. This analysis reveals that the retailer's profit trajectory is non-monotonic, affected by the privacy cost difference. We then explore pricing and profit variations in a different model where retailers solely use personalized recommendations, uncovering additional e-commerce strategic intricacies.

### 4.3. Model BB: With personalized recommendation and BBP

By incorporating personalized recommendation and BBP for both new and existing consumers, we explore how the equilibrium outcome changes when consumers decide to share their privacy.

**The Second Period**. We use $x_1$ to denote the consumer's indifference point between retailers A and B. Then we can have

$$v - p_{ao} - x_1 + \beta r_a - c_a = v - p_{bn} - (1 - x_1) + r_b - c_b, \tag{13}$$

$$x_1 = \frac{(\beta r_a - r_b) + p_{bn} - p_{ao} + 1 + c_\Delta}{2}. \tag{14}$$

We place the consumer utility obtained from repeated purchases of retailer A's product on the left side of Eq. (13). Meanwhile, since retailers choose BBP model, existing consumers receive differentiated price $p_{ao}$ offered by retailer A. The right side of Eq. (13) shows the utility of a first-time purchase from retailer B, which sets a differentiated price $p_{bn}$ to the new consumer. In addition, $x_2$ denotes the point at which the consumer is indifferent between retailers A and B. Then, there are the following equations:

$$v - p_{an} - x_2 + r_a - c_a = v - p_{bo} - (1 - x_2) + \beta r_b - c_b, \tag{15}$$

$$x_2 = \frac{(r_a - \beta r_b) + p_{bo} - p_{an} + 1 + c_\Delta}{2}. \tag{16}$$

Both sides of Eq. (15) represent the utility of the consumer choosing retailers A and B respectively. Then the profit functions in the second period are $\pi_{a2} = p_{ao}x_1 + p_{an}(x_2 - x_n) - cr_a$ and $\pi_{b2} = p_{bo}(1 - x_2) + p_{bn}(x_n - x_1) - cr_b$.

**The First Period**. Online retailers sell uniformly priced products to all consumers in the first period. Considering the utility in the second period, $x_n$ denotes that the marginal consumer is indifferent between purchasing the product from retailers A or B. Then we get

$$v - p_{a1} - x_n + r_a - c_a + [v - p_{bn}^{e*} - (1 - x_n) + r_b - c_b] = v - p_{b1} - (1 - x_n) + r_b - c_b + [v - p_{an}^{e*} - x_n + r_a - c_a] \tag{17}$$

The consumer's utility from purchasing the product of retailer A in the first period is on the left side of Eq. (17), while the utility obtained by purchasing retailer B's product is represented in the right side of Eq. (17). The pricing superscript $e*$ denotes the consumer's expectation of the new consumer's price in period 2. Then, we can derive the indifference point $x_n$ in the first period with respect to $p_{a1}$ and $p_{b1}$ in Eq. (18).

$$x_n = \frac{4 + 2c_\Delta - 3p_{a1} + 3p_{b1} + (1+\beta)r_a - (1+\beta)r_b}{8}. \tag{18}$$

The online retailers set the prices in the first period to maximize the total profit of the two periods. Then the profit functions are

$$\pi_{a1} = p_{a1}x_n + \pi_{a2}^{e*}, \tag{19}$$

$$\pi_{b1} = p_{b1}(1 - x_n) + \pi_{b2}^{e*}. \tag{20}$$

Moreover, there exists pure strategy equilibrium when online retailers adopt both personalized recommendation and BBP. We use first-order conditions to solve the prices in the first period.

In the following analysis, we conduct a comparative examination of prices across the two distinct periods specified in Model BB. This investigation aims to explore the retailer's pricing strategy, with a particular focus on any potential price discrimination between new and existing consumers.

**Corollary 2.** *In Model BB, when retailers make personalized recommendation and BBP, we obtain*

*(i) The prices in period 2 are always lower than those in period 1.*
*(ii) For retailer A(B), if $c_\Delta < \tilde{c}_5(\tilde{c}_6)$, the price of new consumers is higher than that of existing consumers. Otherwise, the price of new consumer is lower than that of existing consumers.*

Corollary 2 shows that retailers use BBP to lower prices in the second period, capitalizing on consumers' reduced price sensitivity during the initial BBP-absent purchase. Retailers initially increase prices to gain profits, planning for future discounts. Later, to retain existing consumers and attract new ones, they offer lower prices, especially for new consumers seeking higher recommendation service level at a reduced cost, maximizing market share and profits. Further analysis shows that the price difference between new and existing consumers is influenced by the privacy cost difference. When $c_\Delta$ is low, existing consumers get lower prices as the personalization impact is less for repeat buyers, prompting retailers to focus on retaining them. However, as $c_\Delta$ rises, new consumers become more privacy cost-sensitive, driving competitors to lower prices for them to build market demand and increase profits.

## 5. Models comparison

In this section, we analyze the optimal strategic decisions made by online retailers under various models. By comparing the equilibrium prices and profits across these models, we uncover several key insights. Specifically, a comparison between models BN and NN reveals the following findings. In terms of pricing strategy, when retailers implement personalized recommendations, they tend to set lower prices compared to non-personalized modes. This strategy stems from the ability of personalized recommendations to enhance consumer purchase intentions, despite higher recommendation costs. However, as recommendation costs rise, especially with higher $\beta$ values, retailers may need to increase retail prices to prevent further profit declines in subsequent periods, offsetting the adverse impact of recommendation costs.

**Proposition 1.** *In period 2, we find the following results of retailers' prices and profits.*

*(i) The prices in Model BN are always lower than those in Model NN.*
*(ii) For retailer A(B), if $c < \hat{c}_1(\bar{c}_1)$, the profit in Model BN is higher than that in Model NN. Otherwise, the profit in Model BN is lower than that in Model NN.*

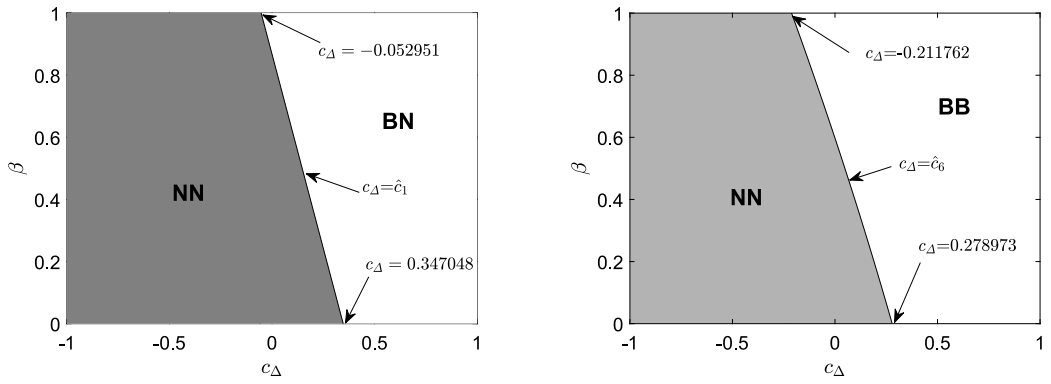
Proposition 1 (ii) indicates that in the second period, retailers implementing personalized recommendations can achieve higher profits when recommendation costs are lower. This is because personalized recommendations can effectively attract and retain consumers, particularly under lower privacy costs. However, with increasing recommendation costs, especially with larger $\beta$ values, recommendation costs are more dominant than the recommendation effect. This principle is exemplified by ANTA's online flagship store, where escalating recommendation costs have diminished ANTA's profits from personalized recommendations. Therefore, it can be concluded that as recommendation costs increase, retailer profits tend to decrease.

Fig. 2 (a) visually illustrates the optimal strategy ranges of NN and BN models, highlighting the influence of $\beta$ and $c_\Delta$ on these ranges. We find that when $c_\Delta$ exceeds the threshold $\hat{c}_1$, the BN model becomes the optimal strategy as it effectively leverages personalized recommendations to attract consumers. However, when $c_\Delta < c < \hat{c}_1$, the NN model emerges as the optimal strategy by avoiding additional costs associated with personalized recommendations. Thus, the values of $c_\Delta$ and $\beta$ significantly impact retailers' optimal strategy choices.

**Proposition 2.** *The comparison of the results for the first period of Models BN and NN are following:*

*(i) For retailer A(B), if $c_\Delta < \tilde{c}_7(\tilde{c}_8)$, the price in Model BN is lower than that in Model NN. Otherwise, the price in Model BN is higher than that in Model NN.*
*(ii) For retailer A(B), if $c_\Delta < \hat{c}_2(\hat{c}_3)$, the profit in Model BN is lower than that in Model NN. Otherwise, the profit in Model BN is higher than that in Model NN.*

By comparing the first period of Models BN and NN, we draw the following conclusions. When $c_\Delta$ is small, the retailer adopting personalized recommendation sets a lower price than the retailer without personalized recommendation. This is because when consumers share their privacy, retailers can distinguish between new and existing consumers based on their personal data. When the privacy cost difference is small, consumers are less sensitive to privacy cost, allowing retailers without personalized recommendation to increase their prices. As $c_\Delta$ increases, retailers enhance




(a) Comparison of Models NN and BN. (b) Comparison of Models NN and BB.

**Fig. 2.** Profit comparison and optimal range.

the level of personalized recommendation to stimulate demand. At this time, the retailer who adopts personalized recommendation raises the prices appropriately to compensate for the loss incurred by the recommendation cost.

From a profit perspective, Proposition 2 (*ii*) shows that the retailer who chooses personalized recommendation obtains a higher profit when $c_\Delta$ is large, highlighting the significance of personalized recommendation strategy. The reason for this result is that retailers use personalized recommendation to attract consumers and thus expands the consumer market, resulting in higher profit compared to the retailer not adopting personalized recommendation when the privacy cost difference is large. In addition, the results reveal the relationship between personalized recommendation and privacy cost. Consumer utility derived from consumption can be simplified as the recommendation utility minus the privacy cost difference ($r_i - c_\Delta$). Therefore, within a certain range, it is beneficial for retailers to choose personalized recommendation.

Expanding on the preceding discussion, we now delve into the impact of retailers adopting BBP on prices and profits, while considering the influence of $c_\Delta$.

**Proposition 3.** *By comparing the equilibrium results of Model BN and BB, we have the following results.*

*(i) If $c_\Delta > \bar{c}_3(\bar{c}_5)$, retailer A(B)'s price for existing consumers in Model BB is higher(lower) than that in Model BN. Otherwise, the price in Model BB is lower(higher) than that in Model BN.*

*(ii) If $c_\Delta > \bar{c}_4(\bar{c}_6)$, retailer A(B)'s price for new consumers in Model BB is lower(higher) than that in Model BN. Otherwise, the price in Model BB is higher(lower) than that in Model BN.*

*(iii) If $c_\Delta < \hat{c}_4(\hat{c}_5)$, retailer A(B)'s profit in Model BB is higher than that in Model BN. Otherwise, the price in Model BB is higher than that in Model BN.*

Proposition 3 shows that when the retailer chooses BBP and the privacy cost difference is large, retailer A charges a higher price for existing consumers and a lower price for new consumer s. Under the same conditions, retailer B's prices for new and existing consumers are exactly the opposite of retailer A's prices. These results may seem counterintuitive. The reason is that retailers' pricing decisions are affected by BBP. In a competitive market, to maintain their existing consumer base and capture more market share, retailers lower prices for existing consumers in the second period. Simultaneously, retailers expand demand by increasing the recommendation service level for new consumers, and the higher recommendation cost results in higher prices for new consumers.

When comparing the equilibrium results of Models BN and BB, it is observed that with a larger value of $c_\Delta$, the profit of the online retailer utilizing only personalized recommendation is lower compared to that of the retailer implementing personalized recommendation and BBP. Online retailers leverage privacy information provided by consumers to differentiate between new and existing consumers. When employing BBP, retailers charge higher prices to new consumers, consequently yielding higher profits. Even if lower prices are offered to existing consumers, this approach tends to increase profits in the second period. Hence, the practice of higher pricing by retailers facilitates higher profits, thereby giving them an advantage in selecting personalized recommendation and BBP over relying solely on personalized recommendation. Considering the influence of BBP, we subsequently analyze the price and profit dynamics in the first period.

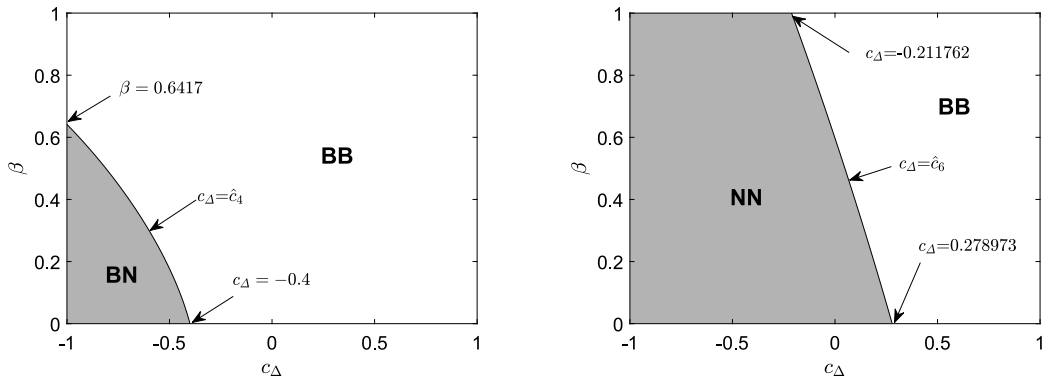
**Proposition 4.** *In the first period, we have the following results:*

*(i) The prices in Model BB are always higher than those in Model BN.*

*(ii) The profits in Model BB are always higher than those in Model BN.*

In Proposition 4, we compare the first-period equilibrium results between Models BN and BB. The analysis reveals that retailer employing both personalized recommendation and BBP tends to set higher prices. This is due to two primary factors. Firstly, by tailoring their recommendation to align with consumers' preferences, retailers enhance the appeal of their product and generate greater demand. Secondly, retailer utilizing BBP must moderate price competition in the initial period, considering that they lower their prices in the subsequent period due to the intensifying competition. Consequently, to optimize their profit, they set higher prices in the first period.

Proposition 4 (*ii*) shows that retailers are more profitable under Model BB compared to Model BN when $c_\Delta$ is large. The reason is that the retailer can set personalized prices for new and existing consumers when choosing BBP, which attracts more consumers to buy the product. When retailer who chooses personalized recommendation and BBP charges higher prices, their profits increase with higher pricing. Then we analyze the optimal decision range when retailers use personalized recommendation and BBP. From Fig. 2(a), it can be discovered that when $c_\Delta > \hat{c}_4$, retailers choose BBP as the better strategy, i.e., Model BB is the optimal strategy. When $c_\Delta < \hat{c}_4$, Model BN is the optimal strategy.




(a) Comparison of Models BN and BB. (b) The optimal strategy.

**Fig. 3.** Profit comparison and optimal strategy.

In the subsequent section, we compare the equilibrium prices and profits of retailers opting for personalized recommendation and BBP. This comparison extends the analysis presented in Proposition 4, which focuses on the equilibrium results in the initial period for Models BN and BB (see Fig. 3).

**Proposition 5.** *In the second period, we compare Model NN and BB and find the following results.*

*(i) If $c_\Delta > \tilde{c}_9(\tilde{c}_{10})$, retailer A(B)'s price for existing consumers in Model BB is higher(lower) than that in Model NN. Otherwise, the price in Model BB is lower(higher) than that in Model NN.*
*(ii) The prices for new consumers in Model BB are always lower than those in Model NN.*
*(iii) If $c_\Delta > \bar{c}_4(\hat{c}_6)$, retailer A(B)'s profit in Model BB is higher than that in Model NN. Otherwise, the price in Model BB is higher than that in Model NN.*

In Proposition 5, we compare the equilibrium outcomes between Models NN and BB. Our analysis reveals that when the privacy cost difference is substantial, the price charged by retailer A employing BBP exceeds that in Model NN. This phenomenon can be due to several key factors. Firstly, following the first period, consumers become more cautious in their decisions during the second period. In Model BB, retailer B strategically lowers prices for existing consumers to maintain market competitiveness. Conversely, retailer A opts to enhance the recommendation service level, thereby improving the service experience for regular consumers, which incurs higher cost. Consequently, retailer A sets a higher price. These contrasting strategies highlight the various approaches taken by different retailers in managing consumer information. Furthermore, for retailers utilizing BBP, the price charged to new consumers consistently remains higher. This observation indicates that retailers prioritize investing recommendation costs in acquiring new consumers, and they are not affected by other factors in this regard.

In addition, employing BBP with an increasing price for new consumers enables, the retailer to achieve higher profit in the second period. The research shows that the profit of the retailer may be higher when not implementing personalized recommendation and BBP than using both personalized recommendation and BBP. This is because when the retailer tries to actively improve the profit by using personalized recommendation and BBP, it also entails an increase in recommendation cost. This leads to a decrease in the profit of the retailer. Contrary to their expectations. A more intuitive analysis from Fig. 2(b) shows that Model NN is chosen as optimal when $c_\Delta < \hat{c}_6$. When the parameter ranges are reversed, Model BB becomes optimal. We find that Model BB occupies a larger region, suggesting that it is more favorable for the industry to opt for BBP.

**Proposition 6.** *In the first period, we have the following results.*

*(i) For retailer A(B), if $c_\Delta > \hat{c}_7(\hat{c}_8)$, the price in Model BB is higher than that in Model NN. Otherwise, the price in Model BB is lower than that in Model NN.*
*(ii) For retailer A(B), if $c_\Delta > \bar{c}_5(\bar{c}_6)$, the profit in Model BB is higher than that in Model NN. Otherwise, the profit in Model BB is lower than that in Model NN.*

Proposition 6 states that in the first period of Model BB, retailer A, who chooses personalized recommendation and BBP, sets higher price to consumers when $c_\Delta > \hat{c}_7$. When $c_\Delta > \hat{c}_8$, retailer B sets higher price. However, this result is reversed if the parameter range changes. The reason for retailers to set high prices is to increase their profits in the first period. In this scenario, personalized recommendation and BBP intensify price competition among retailers. To compensate for the loss of profit due to competition, retailers raise their prices in the first period.

In addition, when $c_\Delta$ falls within a substantial interval, retailer A experiences considerable profit, while retailer B encounters the opposite outcome. Subsequently, we delve into the underlying reasons for these contrasting results from the perspective of retailer profits. Firstly, the retailers maximize the total profit accruing in two distinct periods, and the profit fluctuations during these periods are influenced by competitors' actions. Secondly, the retailer's profit undergoes changes as they adopt different pricing strategies. Additionally, the implementation of more intricate strategic measures, such as personalized recommendation and BBP, can sometimes yield positive outcomes for retailers. Furthermore, the aforementioned results demonstrate that retailers employing identical strategies may obtain contrasting outcomes.

In the following section, we examine the effect of the specific factors, namely the recommendation service level and privacy cost difference that influence retailers' strategies.

## 6. Sensitivity analysis and main results

In this section, we draw the conclusions from the sensitivity analysis conducted on the two factors of recommendation service level and privacy cost difference. This analysis enables us to understand the optimal strategy adopted by retailers.

### 6.1. Recommendation service level

In Model BN, retailers introduce personalized recommendation in the second period. In Model BB, retailers also introduce BBP based on personalized recommendation. By analyzing the two main models, we find that the recommendation service level plays an important role in the decision-making process.

**Corollary 3.** *In the second period of Model BN, we have the following results.*

*(i) The prices are constantly increasing with the privacy cost difference.*
*(ii) If $r_a > \tilde{r}_1(\tilde{r}_2)$, retailer A(B)'s profit increases with the privacy cost difference. Otherwise, the profits decrease with the privacy cost difference.*

Corollary 3 highlights that when retailers do not choose behavior-based price discrimination, they mainly rely on the consumption effect of personalized recommendation to compete in the market and gain market share. At first, in the second period of Model BN, as retailers' recommendation service level continues to increase, their prices also rise. As the level of recommendation continues to increase, retailers can provide better product service and experience to their consumers. In this regard, personalized recommendation can expand the market demand by catering to consumers needs and reducing the time spent in the purchasing process. As retailers' recommendation cost increases, they seek to minimize the loss from recommendation cost by increasing the prices. In addition, in the second period, when the level of recommendation is high, retailers' profits increase. When the recommendation service level is low, retailer profits do not increase with the increase in the recommendation service level, this is because the retailers benefit brought by the recommendation service level increase cannot be covered by the cost rise. As the level of recommendation increases, the effect on consumers is more significant and retailers are able to expand consumer market demand. Therefore, due to the increase in the cost of personalized recommendation, retailers maximize profits by increasing prices in the second period. From Corollary 4, the price and profit of the retailer in the first period are also affected by personalized recommendation as elaborated below.

**Corollary 4.** *In Model BN, we also have the following results in the first period.*

*(i) The prices are constantly increasing with the privacy cost difference.*
*(ii) If $r_a > \tilde{r}_3(\tilde{r}_4)$, retailer A(B)'s profit increases with the privacy cost difference. Otherwise, the profits decrease with the privacy cost difference.*

Initially, in the first period of Model BN, price level rises within a critical range as retailers' recommendation service level continues to increase. In the first period, retailers' price changes remain monotonous due to the increase in recommendation cost, prompting them to raise their prices to offset the negative impact of recommendation cost. Then, at higher recommendation service level, retailers' profits increase as the recommendation service level increases. When recommendation service level is low, retailers' profits do not increase with the recommendation service level. This is because low recommendation service level hinder retailers from attracting more consumers. As the recommendation cost surpasses the benefit of price increases, this change causes retailers' profits to decrease as the recommendation service level increases. Retailers adjust the recommendation service level downwards to slow down the growth rate of recommendation costs. As a result, retailers' first-period profits are likely to rise even if they raise prices at the time of recommendation. In Model BB, retailers choose both personalized recommendation and BBP strategies. We analyze the impact of the recommendation service level in the following corollary.

**Corollary 5.** *After analyzing the impact of the recommendation service level, we have the following results in Model BB.*

*(i) The prices are constantly increasing with the privacy cost difference.*
*(ii) If $r_a > \tilde{r}_5(\tilde{r}_6)$, retailer A(B)'s profit increases with the privacy cost difference. Otherwise, the profits decrease with the privacy cost difference.*

It follows from Corollary 5 that as the recommendation service level increases over time, retailers increase their prices for existing consumers. Traditionally, it is suggested that prices for new consumers should be lowered. New consumers are more sensitive to retailer's recommendation service level. Retailers reduce prices to stimulate the purchasing power of new consumers to compensate for the impact of recommendation service levels on new customers. However, Corollary 5 presents a counterintuitive conclusion that retailers' pricing for new consumers increases with recommendation service level. This is because retailers opting for BBP incur higher cost and increase their prices to minimize losses. From a profit perspective, retailers' revenues increase with progressively higher levels of recommendation. This implies that at higher recommendation service level, the change is beneficial to retailers regardless of whether they reduce their prices or not, and the increase in their profits outweighs the potential loss incurred by new consumers during their purchases.

### 6.2. Privacy cost difference

The preceding section describes the effect of recommendation service level on retailers' decisions. To understand how the privacy cost difference ($c_\Delta$) influences retailers' optimal strategies, we examine the effects of privacy cost difference on retailers' prices and profits.

**Corollary 6.** *In Model BN, the role of privacy cost difference is analyzed below.*

*(i) For retailer A(B), the price increases(decreases) with the privacy cost difference.*
*(ii) If $c_\Delta > \hat{c}_{\Delta 1}(\hat{c}_{\Delta 2})$, retailer A(B)'s profit increases with the privacy cost difference. Otherwise, the profits decrease with the privacy cost difference.*

When consumers choose to disclose private information, the pricing strategies of retailers A and B who choose to personalize their recommendation exhibit different trends as $c_\Delta$ changes gradually. As the difference in retailers' privacy cost increases, retailer A's price increases while retailer B's price decreases. The reason for this is that as $c_\Delta$ grows, consumers must decide which retailer to buy from, leading to intensified competition among retailers. Retailers gain market share by offering higher levels of service or lower prices, reflecting the two competitive strategies adopted by retailers A and B. In addition, when $c_\Delta$ is large, retailers' profits increase with $c_\Delta$. When $c_\Delta$ is smaller than this threshold, retailers' profits are not expected to increase. Furthermore, retailer B attracts consumers by lowering its price, a strategy that helps expand the consumer market and increases its profits in the second period. We find that as privacy cost increases, consumers become more cautious in their choices, thereby influencing retailers' pricing decisions and second-period revenues. In summary, we examine how privacy cost difference in the first period affect retailers who solely opt for personalized recommendation.

**Corollary 7.** *In the first period, we also get the following results.*

*(i) For retailer A(B), the price increases(decreases) with the privacy cost difference.*
*(ii) For retailer A(B), the profit increases(decreases) with the privacy cost difference.*

Corollary 7 shows that in the first period, retailers can adopt personalized recommendation when consumers choose to disclose private information. As $c_\Delta$ changes gradually, the pricing and profit changes for retailers A and B are different. As $c_\Delta$ increases, retailer A's price increases, while retailer B's price decreases. We find that retailer A's profit increases as $c_\Delta$ increases. Conversely, retailer B's profit decreases as $c_\Delta$ increases. This finding indicates that the privacy cost is lower when consumers buy from Retailer A in the first period. Even if retailer A raises its price, consumers are still likely to choose to buy from retailer A. As a result, the market expansion contributes to the retailer's profitability. Since retailer B requires consumers to pay a higher privacy cost, the price has to be lowered to expand consumer demand. However, consumers are sensitive to privacy cost, which results in retailer B being unable to attract more consumers, and its profitability decreases with lower prices. Next, we analyze the impact of changes in privacy cost difference on retailers who choose BBP.

**Corollary 8.** *In Model BB, we have the results as follows.*

*(i) The prices of retailer A(B) for new and existing consumers increase(decrease) with the privacy cost difference.*
*(ii) If $c_\Delta > \hat{c}_{\Delta 3}(\hat{c}_{\Delta 4})$, retailer A(B)'s profit increases with the privacy cost difference. Otherwise, the profits decrease with the privacy cost difference.*

As shown in Corollary 8, the price charged by retailer B decreases as $c_\Delta$ increases. However, retailer A's price raises as $c_\Delta$ increases. Regardless of how $c_\Delta$ changes, retailers remain consistent in their attitude toward new and existing consumers. For example, when Adidas and Nike brand stores compete online, Adidas decides to lower price based on the purchasing behavior of consumers, but Nike decides to increase the price. Moreover, we obtain an interesting conclusion. As $c_\Delta$ gradually increases, the retailer employing personalized recommendation and BBP does not necessarily set lower price to keep consumers. Even if Nike sets higher price, consumers repurchase from Nike due to their better experience in their first purchase. However, a different scenario can emerge. In a competitive market, consumers who face increasing privacy cost difference choose to purchase more cautiously. Consumers' privacy concerns intensify the competition in the market, resulting in the retailer to reduce price to attract consumers as $c_\Delta$ increases. Observing from profit perspective, when $c_\Delta$ increases, retailers' profits increase accordingly. This is because consumers are less attracted to lower prices, and retailers can gain significant profits from the price increases. Thus, retailers' profits increase as demand increases. As the extension to the above findings, we investigate the retailer's optimal strategy choices through numerical analysis.

### 6.3. The strategy choice

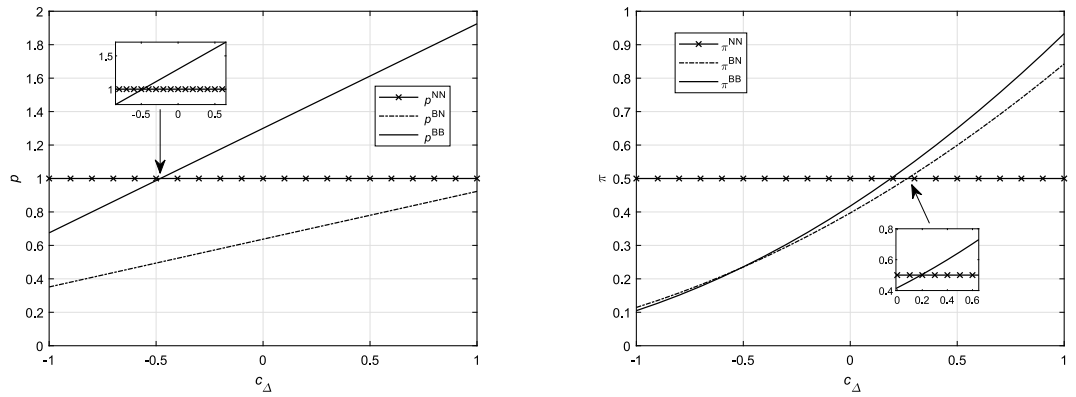
Our analysis of recommendation service level and privacy cost differences highlights a key consideration for retailers choosing between personalized recommendations and BBP. It examines whether retailers should use both strategies or focus solely on personalized recommendations. Proposition 7 outlines the optimal strategy for retailers through numerical comparisons, providing insights into pricing and profit maximization.

**Proposition 7.** *We compare the prices and profits of three models and obtain the following results.*

*(i) If $c_\Delta < \hat{c}_{\Delta 5}$, retailers' prices in Model NN are higher than those in Models BN and BB. If $c_\Delta > \hat{c}_{\Delta 5}$, retailers' prices in Model BB are higher than those in Models NN and BN.*
*(ii) If $c_\Delta < \hat{c}_{\Delta 6}$, the profits of retailers in Model NN are higher than those in Models BN and BB. If $c_\Delta > \hat{c}_{\Delta 6}$, the profits of retailers in Model BB are higher than those in Models NN and BN.*
*(iii) If $c_\Delta < \hat{c}_6$, neither retailer adopts personalized recommendation nor BBP. If $c_\Delta > \hat{c}_6$, both retailer adopts personalized recommendation and BBP.*

Proposition 7 compares prices among models NN, BN, and BB. When $c_\Delta < \hat{c}_{\Delta 5}$, Model NN has higher prices, indicating that smaller privacy cost differences drive competition and reduce prices with personalized recommendations and BBP. Conversely, when $c_\Delta > \hat{c}_{\Delta 5}$, Model BB has the highest prices. As retailers invest in data and personalized strategies, prices in Models BB and BN rise, with Model BB always having higher profit than Model BN, suggesting that personalized recommendations and BBP are optimal. Graphical analysis confirms these findings, showing that as $c_\Delta$ increases, prices rise, with Model BN generally having lower prices(Fig. 4). Model NN's prices remain stable without personalized recommendations and BBP, supporting our conclusions, and retailer profit distribution is also examined.

We compare profits across three models and find that when $c_\Delta < \hat{c}_{\Delta 6}$, Model NN is more profitable, suggesting that low privacy cost difference favor not using personalized recommendations and BBP. This approach can reduce privacy risks and maintain customer loyalty if retailers lack the capability for these strategies. Conversely, when $c_\Delta > \hat{c}_{\Delta 6}$, Model BB is more profitable, while Model NN is the least profitable. Higher privacy cost difference benefit retailers who use personalized recommendations and BBP, as these strategies attract more consumers and increase revenue. Fig. 4(a) illustrates that as $c_\Delta$ increases, profits rise for retailers using personalized recommendations and BBP, whereas Model NN's pricing remains stable. Fig. 4(b) shows that, at moderate values, retailers with personalized recommendations and BBP earn higher profits. Based on this, the optimal strategies are either Models BB or NN. For low privacy cost difference, avoiding personalized recommendations and price discrimination is best, while with higher costs, using personalized recommendations and BBP is optimal.




(a) The comparison of online retailer price. (b) The comparison of online retailer profit.

**Fig. 4.** Comparison of online retailers' equilibrium outcomes($r_a = 0.3, r_b = 0.5, \beta = 0.5$).

## 7. Extensions

When considering personalized recommendation and behavior-based pricing strategies, a cost-oriented examination of privacy concerns is crucial. The previous sections provide insights into how retailers should develop their pricing and personalized recommendation strategies while accounting for privacy costs. In the extension, we further explore retailers strategically decision when considering various consumer sensitivity to privacy and decision process.

### *7.1. Privacy-sensitive and non-privacy-sensitive consumers*

This section extends the previous studies by incorporating both privacy-sensitive and non-privacy-sensitive consumers. So as to analyze how the retailers' pricing strategies and profits are affected by factors such as the level of recommendation and the cost of privacy. To examine the effect, we introduce a parameter $\lambda$ $(0 < \lambda < 1)$, designed to measure size of consumers with different sensitivity to privacy issues. Through this parameter, we categorize consumers into different groups based on varying degrees of privacy concerns, thereby reflecting the acceptance and effectiveness of personalized recommendation strategies across different consumer segments. Specifically, the parameter $(\lambda)$ allows us to identify consumers highly concerned with privacy protection, as well as those who may be less hesitant to share personal information in exchange for personalized services. This differentiation enables us to analyze more precisely the utility of personalized recommendation strategies across different levels of privacy sensitivity and how it impacts consumer purchasing decisions and retailers' market strategies.

When consumers are privacy-sensitive and opt out of privacy options, retailers cannot utilize personalized recommendation and BBP. However, if consumers are non-privacy-sensitive, they do not opt out of privacy options, allowing retailers to employ personalized recommendation and BBP strategies. First, we examine the retail model focusing solely on personalized recommendation (Model BN). In the second period, we denote the marginal consumer's indifference point $x_m = \frac{p_{b2}-p_{a2}+1}{2}$ as the choice between retailers A and B, where $\lambda$ represents the size of non-privacy-sensitive consumers and $(1-\lambda)$ represents the size of privacy-sensitive consumers. We have $\pi_{a2} = p_{a2}[\lambda(x_1 + x_2 - x_n) + x_m(1-\lambda)] - cr_a$ and $\pi_{b2} = p_{b2}[\lambda(1-x_2+x_n-x_1)+(1-x_m)(1-\lambda)] - cr_b$. Next, we investigate retailers employing both personalized recommendation and BBP (Model BB) strategies simultaneously. In Model BB, retailers consider not only consumer privacy preferences but also incorporate their purchase histories to determine pricing. This approach enables retailers to predict consumer demand more accurately and adjust personalized recommendation service level and pricing strategically. Similar with the model settings in Section 4, we denote the indifference point $x_m = \frac{1}{2}\left(-p_{\text{an}} + p_{\text{bn}} + 1\right)$ representing marginal consumers' choices between retailers A and B. Consequently, we derive the profit formulas for retailers A and B in the second period. $\pi_{a2} = -cr_a + (1-\lambda)x_m\left(p_{\text{an}} + p_{\text{ao}}\right) + \lambda\left(p_{\text{an}}\left(x_2 - x_n\right) + x_1 p_{\text{ao}}\right)$ and $\pi_{b2} = -cr_b + (1-\lambda)\left(1-x_m\right)\left(p_{\text{bn}} + p_{\text{bo}}\right) + \lambda\left(p_{\text{bn}}\left(x_n - x_1\right) + \left(1-x_2\right)p_{\text{bo}}\right)$.

Through these models, we infer to discuss how retailers formulate effective personalized recommendation and pricing strategies while considering privacy cost.

Firstly, we conduct an analysis of pricing strategies. When personalized recommendation positively influence consumer purchasing behavior, retailer A can compensate the cost of personalized recommendation by increasing prices. For retailer B, irrespective of variations in retailer A's recommendation service level, pricing under model BN consistently exceeds that under model NN. This may be because retailer B can more effectively segment consumer groups under model BN and accordingly set higher prices. Secondly, we analyze retailers' profits. Retailer A consistently earns higher profits under model BN compared to model NN. This is likely because model BN allows retailer A to adjust its strategies based on consumer privacy preferences and recommendation sensitivity, thereby achieving higher profits. For retailer B, its profit performance under model BN depends on recommendation service level and recommendation cost. This suggests that the cost-effectiveness of personalized recommendation is clearer at lower recommendation service level within model BN.

In considering how consumer privacy sensitivity $(\lambda)$ affects retailers, we conduct the following analysis. First, when consumers place a lower value on privacy (i.e., $\lambda$ is small), they may be more willing to accept personalized recommendations, thereby increasing the likelihood that retailers can raise prices to achieve higher profits. On the other hand, consumers with high privacy sensitivity are unwilling to sacrifice privacy for personalized services, which limits the space for retailers to raise profits through personalized recommendations. Additionally, retailers can use information on consumer privacy sensitivity to segment the market and provide more personalized services to consumers who are not sensitive to privacy. Incorporating both privacy-sensitive and non-privacy-sensitive consumers, the extension enriches the previous findings and provides insights for retailers on how to develop effective personalized recommendation and pricing strategies while considering privacy costs. The research emphasizes the importance of balancing personalized services and consumer privacy rights in the digital economy.

### 7.2. Myopic consumers

In the digital age, consumers are more aware than ever of privacy protection, and this heightened awareness is significantly influencing their purchasing decisions and their willingness to accept personalized services. In the previous section of our model, we primarily discussed forward-looking consumers, who anticipate subsequent purchases when initially buying a product or service. In this subsection, we analyze myopic consumers, who also have the opportunity to reassess their privacy policy choices in the initial stages of both the first and second periods. This pattern makes the two periods comparable, and we model it based on the foundational model.

Consumers evaluate and select their privacy strategies during two distinct periods when purchasing products or services. We consider two privacy strategies: privacy protection and non-privacy protection. Privacy-protective consumers prefer that retailers do not use their purchase history for personalized recommendations and behavior-based pricing (BBP). This choice limits the retailer's ability to utilize consumer data, thereby impacting marketing strategies and pricing decisions. Depending on consumer choices, four scenarios emerge. First, consumers protect their privacy in both periods (Model NN). Second, consumers do not protect their privacy in either period, allowing retailers to access privacy information for personalized strategies (Models BN and BB). Third, consumers do not protect their privacy initially but do so in the second period, preventing retailers from implementing personalized strategies in the second period (Model DN). Fourth, consumers protect their privacy initially but not in the second period, allowing retailers to implement personalized strategies in the latter period (Models ND and DD).

Based on Section 4, we investigate the remaining scenarios. Firstly, we analyze the model (Model DN) where retailers are unable to implement personalized strategies in Period 2. In period 2, we denote the indifference point as $x_0 = \frac{p_{b2}-p_{a2}+1}{2}$ to represent marginal consumers' choice between retailers A and B, yielding the profit formulas $\pi_{a2} = p_{a2}x_0$, $\pi_{b2} = p_{b2}(1-x_0)$. In Period 1, we have a new indifference point $x_n = \frac{1}{2}(r_a - p_{a1} - r_b + p_{b1} + c_\Delta + 1)$, along with the profit formulas for retailers $\pi_{a1} = p_{a1}x_n - cr_a$, $\pi_{b1} = p_{b1}(1-x_n) - cr_b$. Next, we analyze the scenario where retailers were unable to provide recommendation in the first period and only engaged in personalized recommendation in the second period. Similarly, in the first period, we denote the indifferent point of $x_0 = \frac{p_{b1}-p_{a1}+1}{2}$ for marginal consumers choosing between retailers A and B, and derive the corresponding profit formulas $\pi_{a1} = p_{a1}x_0$ and $\pi_{b1} = p_{b1}(1-x_0)$.

Based on these descriptions, we can draw the following conclusions regarding personalized strategies for myopic consumers. For myopic consumers, firstly, they place significant emphasis on protecting personal data, especially concerning purchase history and personalized recommendations. Privacy-conscious consumers wish to minimize retailers' use of their data for personalized recommendations and dynamic pricing. Secondly, some consumers may have varying privacy preferences at different times; they may opt to protect their privacy at certain times while being less concerned at others. Regarding personalized strategies for retailers, we propose the following recommendations. Retailers should respect consumers' privacy preferences and strictly adhere to privacy protection policies during data collection and usage, refraining from unauthorized use of consumer data for personalized recommendations or dynamic pricing. Furthermore, for consumers who exhibit varying privacy preferences over time, retailers need to design flexible marketing strategies. For instance, when consumers choose to protect their privacy, retailers should provide generalized advertisements and product recommendations. Conversely, when consumers agree to use their data, personalized recommendations and discounts can be offered. By understanding consumers' privacy preferences and behavioral consistencies, retailers can effectively formulate personalized strategies while maintaining a balance between compliance and consumer trust, thereby enhancing market competitiveness and consumer satisfaction.

## 8. Management significance

In this section, we discuss the main findings and explain the underlying reasons for these results and their managerial applications.

The optimal strategy for online retailers is to either adopt both personalized recommendations and BBP or to avoid both. This recommendation is based on consumer preferences and privacy concerns. When consumers prioritize privacy, avoiding both methods allows retailers to lower costs and achieve higher profits. On the other hand, using only personalized recommendations can increase costs and decrease profits [5]. Combining both methods can provide a competitive edge and achieves Pareto optimality, as evidenced by FENDI's 16 percent increase in net profit through differentiated pricing. Therefore, the optimal strategy is to adopt both strategies simultaneously or neither.

Our analysis reveals that retailers employ a strategic pricing approach by raising prices in the first period and then lowering them in the second to stimulate demand. In the second period, where personalized recommendations and differential pricing are utilized, retailers encounter higher costs and greater consumer sensitivity to privacy. Thus, they reduce prices later to attract consumers, balancing out the higher profits gained initially. For example, Nike cuts the price of slow-selling shoes by 27 percent on repeat purchases, leading to a 12 percent increase in net profits. Retailers using only personalized recommendations often maintain low prices throughout due to the method's limited capability to drive demand [50]. Therefore, optimal pricing and profits for online retailers are contingent on their chosen strategies.

Furthermore, our results shows that higher recommendation service levels and greater privacy cost differences generally benefit retailers. Improved recommendation services make consumer targeting more precise, which expands market demand and increasing profits. As service levels improve, consumers often willing to accept higher privacy costs for a better shopping experience, allowing retailers to generate extra revenue from consumer information. Additionally, greater privacy cost differences intensify market competition, enabling retailers to leverage differential pricing and Big Data for a competitive edge. For instance, Wish adjusts its strategies based on privacy cost differences compared to other platforms, highlighting the strategic significance of these factors for online retailers.

This study delves into the impacts of personalized recommendation and BBP strategies on business operations and consumer privacy, offering the following managerial insights. First, the research provides a deep understanding of the cost-effectiveness of personalized recommendation and BBP strategies, guiding management in making more informed decisions when developing and implementing their strategies. This insight helps businesses enhance market competitiveness and achieve long-term sustainable development without compromising consumer trust. Second, our findings underscore the potential of personalized strategies in improving operational efficiency. By refining recommendation and pricing models, businesses can better align with consumer demands while respecting consumer privacy rights. This optimization not only improves consumer satisfaction but also ensures that privacy concerns are effectively managed. Such operations not only enhance customer loyalty but also increase the responsiveness of enterprises. Third, our study's insights enable marketing teams to craft more compelling campaigns using personalized recommendation and BBP strategies. These strategies assist businesses in effectively targeting their markets and enhancing the conversion rates of marketing activities. Lastly, in the context of increasingly stringent data privacy and consumer protection regulations, this research offers critical insights into enterprise risk management and compliance. By accurately assessing the costs and benefits of personalized recommendation, businesses can better balance innovation with risk, ensuring the effectiveness of their strategic planning. Overall, this study provides a comprehensive framework for businesses to assess and implement personalized recommendation and BBP strategies, enhancing market competitiveness and promoting sustainable growth while protecting consumer privacy.

## 9. Conclusion

The contemporary landscape reveals an increasing trend emphasis on privacy in the context of personalized recommendations and price discrimination. As consumers become more focused on protecting their personal information, their commitment to privacy is growing. To address these concerns, individuals can now choose to safeguard their privacy when purchasing products or services online through various policies. Academic research primarily focuses on investigating retailers' personalized recommendation systems and price discrimination, while highlighting the influence of privacy concerns on retailer performance [32]. Our study builds on this by examining how consumer privacy concerns influence retailers' strategy on personalized recommendation and price discrimination. Furthermore, we provide both theoretical and managerial insights into the effectiveness of two-period strategies involving personalized recommendations and BBP.

### *9.1. How do retailers choose personalized recommendation and BBP strategies?*

When selecting personalized recommendation and BBP strategies, retailers need to carefully evaluate several factors. Research indicated that relying on a single approach is insufficient; instead, retailers should tailor their strategies to their specific context. Firstly, taking privacy concerns into account, if the privacy cost difference for retailers is minimal, the optimal strategy would be to abstain from implementing personalized recommendation and BBP strategies. This strategy allows retailers to reduce investment cost while maintaining a stable market position. By doing so, retailers can effectively balance meeting consumer privacy protection needs and increasing profits, thus achieving a win-win situation. Secondly, market competition plays a significant role. If there is a significant gap in privacy cost between two retailers, the most favorable course of action is to adopt personalized recommendation and BBP strategies. As the privacy cost difference increases, market competition becomes more intense. In such situations, managers should proactively tackle the challenges and leverage the combined advantages of personalized recommendation and BBP strategies to attract consumers [51].

As exemplified by Target and Louis Vuitton, personalized recommendations can significantly enhance consumers' shopping experiences and increase sales. When privacy cost differentials are minimal, retailers should avoid implementing these strategies to reduce investment costs while maintaining market position. This aligns with Target's cautious stance on privacy protection in consumer data collection. Conversely, when there are significant privacy cost differentials, retailers should leverage personalized recommendations and BBP strategies to attract consumers, similar to Louis Vuitton setting varying prices based on consumer purchasing levels. Therefore, the optimal strategies for online retailers and their competitors involve either adopting both personalized recommendation and BBP strategies or refraining from using either. Additionally, our research reveals that relying solely on personalized recommendations may put online retailers at a competitive disadvantage and struggle to enhance their market competitiveness. To summarize, when making decisions regarding personalized recommendation and BBP strategies, retailers should thoroughly evaluate factors such as privacy cost difference, market competition, consumer sensitivity to privacy, and other relevant considerations. By strategically selecting and integrating various personalized strategies, retailers can effectively address consumer needs and strengthen their market position.

### *9.2. How do online retailers' prices and profits change over two periods under varying strategies?*

We analyzed retailers' prices and profits across various strategies and compared the equilibrium results of online retailers over two periods. Our findings reveal that the optimal pricing and profitability for retailers largely depend on their strategic decisions. Notably, when retailers opt solely for personalized recommendation and adopt uniform pricing, their total profits can improve to some extent. However, this approach results in lower prices and profits compared to the other strategies. In contrast, if retailers choose to utilize both personalized recommendation and BBP simultaneously, their prices and profits can gradually surpass those of the other models. For retailers aiming to maintain stable prices and profits, it is advisable to avoid consumer privacy concerns and prepare for market competition in the future [52].

Our analysis reveals that retailers' optimal pricing and profits are closely tied to their strategic decisions. For instance, while exclusive use of personalized recommendations may boost total profits, it often results in low prices and profits compared to other strategies. This reflects challenges in profit maximization similar to those faced by Target with its personalized product recommendations. Conversely, combining personalized recommendations with BBP strategies can lead to substantial improvements in both prices and profits, akin to Louis Vuitton's strategy of attracting high-end consumers through premium pricing and services. The research findings indicate that the recommendation service level and pricing employed by retailers under different strategies significantly impact overall profit growth. By leveraging these strategies, retailers can better understand consumers' preferences and needs, enabling them to provide more personalized and accurate recommendation. This can enhance consumers' willingness to make purchases and cultivate loyalty, ultimately leading to higher sales and profits. Furthermore, it is crucial for retailers to consider the price elasticity of their products. Products with high price elasticity where consumers are sensitive to price and easily influenced by competitors may benefit from lower prices potentially increasing sales and market share. Therefore, retailers should integrate personalized recommendations with BBP to provide tailored feedback for both new and existing consumers.

### *9.3. How do factors such as the recommendation service level and privacy cost difference influence retailers' strategies?*

Our investigation into the recommendation service level and privacy cost difference on retailers reveals the importance of balancing costs with profits. Both personalized recommendation and BBP strategies require resource and financial investments, such as the developing of recommendation systems, maintaining price differentials, and covering promotional costs. As the recommendation cost increases, the recommendation service level improves incrementally, leading many consumers accept privacy trade-off for enhanced shopping convenience. Retailers must navigate market competition carefully, especially when privacy cost difference varies. Our findings indicate that the level of recommendation and differences in privacy costs significantly impact retailers' strategic choices. As recommendation costs rise, consumers may be more willing to sacrifice privacy for personalized recommendations, demanding retailers to strike a balance between these costs and their profits. This reflects the trade-offs between privacy and convenience faced by companies like Target and Louis Vuitton when collecting and utilizing consumer data. As the privacy cost difference expands, retailers can implement personalized strategies to attract consumer demand and ensure that profits increase in tandem with rising prices [53]. This strategy exemplifies retailers' pursuit of a win-win scenario while addressing privacy concerns. Consequently, despite the

higher cost associated with implementing these strategies, managers can achieve greater profits by adopting the most effective approach in a competitive market.

This article presents several innovative concepts, but also acknowledges certain limitations for future research. Firstly, personalized strategies for products with vertical differentiation and an increased focus on quality discrimination could be considered in future research. In practice, it is also worth considering the situation where certain consumers do not purchase products from online retailers, indicating that the market is not fully covered. Furthermore, there are additional areas worthy of exploration, such as integrating additional channels to optimize personalized recommendation strategies and maintaining long-term consumer engagement. These aspects present valuable opportunities for advancing the field and enhancing the effectiveness of online retail strategies.

## CRediT authorship contribution statement

**Chi Zhou:** Conceptualization, Funding acquisition, Methodology. **Danyang Bai:** Investigation, Writing – original draft. **Tieshan Li:** Formal analysis, Writing – review & editing. **Jing Yu:** Validation.

## Declaration of competing interest

All authors have no conflict of interest.

## Acknowledgments

This work is supported by the Humanity and Social Science Foundation of Ministry of Education of China (No. 23YJA630147), National Natural Science Foundation of China (No. 72301191), Tianjin Philosophy and Social Science Planning Project (No. TJGL22-013).

## Appendix

**Proof of Model NN.** Based on the second-order partial derivatives, we have

$$\frac{\partial^2 \pi_{a2}^{NN}}{\partial {p_{a2}^{NN}}^2} = -1 < 0, \quad \frac{\partial^2 \pi_{a1}^{NN}}{\partial {p_{a1}^{NN}}^2} = -1 < 0.$$

Also,

$$\frac{\partial^2 \pi_{b2}^{NN}}{\partial {p_{b2}^{NN}}^2} = -1 < 0, \quad \frac{\partial^2 \pi_{b1}^{NN}}{\partial {p_{b1}^{NN}}^2} = -1 < 0.$$

It can be obtained that $\pi_{a2}^{NN}$ (or $\pi_{a1}^{NN}$) is a concave function with respect to $p_{a2}^{NN}$ (or $p_{a1}^{NN}$) and there exists a maximum value. Let

$$\frac{\partial \pi_{a1}^{NN}}{\partial p_{a1}^{NN}} = 0, \quad \frac{\partial \pi_{a2}^{NN}}{\partial p_{a2}^{NN}} = 0, \quad \frac{\partial \pi_{b1}^{NN}}{\partial p_{b1}^{NN}} = 0, \quad \frac{\partial \pi_{b2}^{NN}}{\partial p_{b2}^{NN}} = 0,$$

we can easily get the equilibrium results in Model NN are

$$p_{a1}^{NN*} = p_{a2}^{NN*} = p_{b1}^{NN*} = p_{b2}^{NN*} = 1,$$

$$\pi_{a1}^{NN*} = \pi_{a2}^{NN*} = \pi_{b1}^{NN*} = \pi_{b2}^{NN*} = 1/2.$$

**Proof of Model BN.** We obtain that

$$p_{a2}^{BN*} = \frac{7 + 4c_{\Delta} + 2(1+\beta)r_a - 2(1+\beta)r_b}{14}$$

and

$$p_{b2}^{BN*} = \frac{7 - 4c_{\Delta} - 2(1+\beta)r_a + 2(1+\beta)r_b}{14}.$$

Meanwhile, we have $\pi_{a2}^{BN*} = [16c_{\Delta}^2 + 4(1+\beta)^2 r_a^2 + (7 - 2(1+\beta)r_b)^2 + 8c_{\Delta}(7 + 2(1+\beta)r_a - 2(1+\beta)r_b) + 4r_a(7(1 - 7c + \beta) - 2(1+\beta)^2 r_b)]/196$ and $\pi_{b2}^{BN*} = [(-7 + 4c_{\Delta} + 2(1+\beta)r_a)^2 - 4(-7 + 49c - 7\beta + 4(1+\beta)c_{\Delta} + 2(1+\beta)^2 r_a)r_b - 8c_{\Delta}(7 + 2(1+\beta)r_a) + 4(1+\beta)^2 r_b^2]/196$.

Meanwhile, we have

$$\frac{\partial^2 \pi_{a1}^{BN}}{\partial {p_{a1}^{BN}}^2} = \frac{\partial^2 \pi_{b1}^{BN}}{\partial {p_{b1}^{BN}}^2} = -\frac{5}{2} < 0.$$

Let

$$\frac{\partial \pi_{a1}^{BN}}{\partial p_{a1}^{BN}} = \frac{\partial \pi_{b1}^{BN}}{\partial p_{b1}^{BN}} = 0.$$

Then, the equilibrium result of the first period can be obtained as

$$p_{a1}^{BN*} = \frac{14 + 6c_{\Delta} + 3(1+\beta)r_a - 3(1+\beta)r_b}{21},$$

$$p_{b1}^{BN*} = \frac{14 - 6c_\Delta - 3(1+\beta)r_a + 3(1+\beta)r_b}{21}.$$

Also, we find that $\pi_{a1}^{BN*} = [343+72c_\Delta^2+18(1+\beta)^2r_a^2+4c_\Delta(77+18(1+\beta)r_a-18(1+\beta)r_b)+2(1+\beta)r_b(-77+9(1+\beta)r_b)+2r_a(77-294c+77\beta-18(1+\beta)^2r_b)]/588$, $\pi_{b1}^{BN*} = [343 + 72c_\Delta^2 + 18(1+\beta)^2r_a^2 + 4c_\Delta(-77 + 18(1+\beta)r_\Delta - 18(1+\beta)r_b) - 2(1+\beta)r_a(77 + 18(1+\beta)r_b) + 2r_b(77 - 294c + 77\beta + 9(1+\beta)^2r_b)]/588$.

**Proof of Corollary 1.** According to Model NN, we compare the equilibrium results of the two periods in Model BN. We have

$$\pi_{a2}^{BN*} - \pi_{a1}^{BN*} = -\frac{(14 + 6c_\Delta + 3(1+\beta)r_a - 3(1+\beta)r_b)(7 + 2c_\Delta + (1+\beta)r_a - (1+\beta)r_b)}{294}.$$

Let $\pi_{a2}^{BN*} - \pi_{a1}^{BN*} > 0$, we drive $\tilde{c_1} < c_\Delta < \tilde{c_2}$, where

$$\tilde{c_1} = \frac{-7 - r_a - \beta r_a + r_b + \beta r_b}{2}, \quad \tilde{c_2} = \frac{-14 - 3r_a - 3\beta r_a + 3r_b + 3\beta r_b}{6}.$$

While $\pi_{a2}^{BN*} - \pi_{a1}^{BN*} < 0$, we drive $c_\Delta < \tilde{c_1}$ or $c_\Delta > \tilde{c_2}$. Thus, if $\tilde{c_1} < c_\Delta < \tilde{c_2}$, $\pi_{a2}^{BN*} > \pi_{a1}^{BN*}$. Otherwise, $\pi_{a2}^{BN*} < \pi_{a1}^{BN*}$.

Similarly, we can get

$$\pi_{b2}^{BN*} - \pi_{b1}^{BN*} = -\frac{(-14 + 6c_\Delta + 3(1+\beta)r_a - 3(1+\beta)r_b)(-7 + 2c_\Delta + (1+\beta)r_a - (1+\beta)r_b)}{294}.$$

Let $\pi_{b2}^{BN*} - \pi_{b1}^{BN*} > 0$, we drive $\tilde{c_3} < c_\Delta < \tilde{c_4}$, where

$$\tilde{c_3} = \frac{14 - 3r_a - 3\beta r_a + 3r_b + 3\beta r_b}{6}, \quad \tilde{c_4} = \frac{7 - r_a - \beta r_a + r_b + \beta r_b}{2}.$$

In addition, let $\pi_{b2}^{BN*} - \pi_{b1}^{BN*} < 0$, we have $c_\Delta < \tilde{c_3}$ or $c_\Delta > \tilde{c_4}$. So if $\tilde{c_3} < c_\Delta < \tilde{c_4}$, $\pi_{b2}^{BN*} > \pi_{b1}^{BN*}$. Otherwise, $\pi_{b2}^{BN*} < \pi_{b1}^{BN*}$.

From the price dimension, we compare the prices and obtain $p_{a2}^{BN*} - p_{a1}^{BN*} = p_{b2}^{BN*} - p_{b1}^{BN*} = -1/6 < 0$. Therefore, both $p_{a2}^{BN*} < p_{a1}^{BN*}$ and $p_{b2}^{BN*} < p_{b1}^{BN*}$ always hold.

**Proof of Model BB.** Then, we calculate the following equilibrium results of the two periods in Model BB. To ensure that the results are exist, we attempt to find the region in which the Hessian matrix is negative definite. The Hessian matrix can be written as $H_1^{BB}(H_2^{BB}) = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$. $M_1^{BB} = -1 < 0$ and $M_2^{BB} = 1 > 0$. It can be obtained that $\pi_{a2}^{BB}(\pi_{b2}^{BB})$ is a concave function with respect to $p_{ao}^{BB}(p_{an}^{BB})$ and $p_{bo}^{BB}(p_{bn}^{BB})$. There exists a maximum value. Solving $\frac{\partial \pi_{a2}^{BB}}{\partial p_{ao}^{BB}} = 0(\frac{\partial \pi_{a2}^{BB}}{\partial p_{an}^{BB}} = 0)$ and $\frac{\partial \pi_{b2}^{BB}}{\partial p_{bo}^{BB}} = 0$ $(\frac{\partial \pi_{b2}^{BB}}{\partial p_{bn}^{BB}} = 0)$. We can easily have $p_{ao}^{BB*} = \frac{32+18c_\Delta+(1+17\beta)r_a-(17+\beta)r_b}{48}$ and $p_{an}^{BB*} = \frac{8+6c_\Delta-(-7+\beta)r_a+r_b-7\beta r_b}{24}$, $p_{bo}^{BB*} = \frac{32-18c_\Delta-(17+\beta)r_a+r_b+17\beta r_b}{48}$ and $p_{bn}^{BB*} = \frac{8-6c_\Delta+(1-7\beta)r_a-(-7+\beta)r_b}{24}$. We can get the equilibrium profits in the second period are as follows. $\pi_{a2}^{BB*} = \frac{1280+468c_\Delta+(197+\beta(-22+293\beta))r_a^2-512(2+\beta)r_b+(293+\beta(-22+197\beta))r_b^2+12c_\Delta(128+(31+47\beta)r_a-(47+31\beta)r_b)+2r_a(-256(-1+9c-2\beta)+(11+\beta(-490+11\beta))r_b)}{4608}$. $\pi_{b2}^{BB*} = \frac{1280+468c_\Delta^2+(293+\beta(-22+197\beta))r_a^2-512(-1+9c-2\beta)r_b+(197+\beta(-22+293\beta))r_b^2+12c_\Delta(-128+(47+31\beta)r_a-(31+47\beta)r_b)+2r_a(-256(2+\beta)+(11+\beta(-490+11\beta))r_b)}{4608}$.

Following a similar approach, we obtain the equilibrium prices and profits in the first period are $p_{a1}^{BB*} = \frac{32+6c_\Delta+(11-5\beta)r_a+(5-11\beta)r_b}{24}$ and $p_{b1}^{BB*} = \frac{32-6c_\Delta+(5-11\beta)r_a+(11-5\beta)r_b}{24}$. Also, we have $\pi_{a1}^{BB*} = \frac{4352+540c_\Delta^2+(263+\beta(14+263\beta))r_a^2-32(23+55\beta)r_b+(263+\beta(14+263\beta))r_b^2+12c_\Delta(208+45(1+\beta)r_a-45(1+\beta)r_b)+2r_a(880-2304c+368\beta-(7+\beta(526+7\beta))r_b)}{4608}$, $\pi_{b1}^{BB*} = \frac{4352+540c_\Delta^2+(263+\beta(14+263\beta))r_a^2+32(55-144c+23\beta)r_b+(263+\beta(14+263\beta))r_b^2+12c_\Delta(-208+45(1+\beta)r_a-45(1+\beta)r_b)-2r_a(368+880\beta+(7+\beta(526+7\beta))r_b)}{4608}$.

**Proof of Corollary 2.** Based on Model BB, we compare the prices in Model BB. We find $p_{ao}^{BB*} - p_{a1}^{BB*} = \frac{-32+6c_\Delta+3(-7+9\beta)r_a+3(-9+7\beta)r_b}{48} < 0$ always holds. Similarly, we have $p_{an}^{BB*} - p_{a1}^{BB*} = \frac{-6+(-1+\beta)r_a+(-1+\beta)r_b}{6} < 0$ always holds. Therefore, $p_{ao}^{BB*} < p_{a1}^{BB*}$, $p_{an}^{BB*} < p_{a1}^{BB*}$. In a similar way, $p_{bo}^{BB*} - p_{b1}^{BB*} = \frac{-32-6c_\Delta+3(-9+7\beta)r_a+3(-7+9\beta)r_b}{48} < 0$ always holds and $p_{bn}^{BB*} - p_{b1}^{BB*} = \frac{-6+(-1+\beta)r_a+(-1+\beta)r_b}{6} < 0$ always holds. So that, we get $p_{bo}^{BB*} < p_{b1}^{BB*}$ and $p_{bn}^{BB*} < p_{b1}^{BB*}$. By comparing the prices of new and existing consumers, let $p_{an}^{BB*} - p_{ao}^{BB*} = \frac{-16-6c_\Delta+(13-19\beta)r_a+(19-13\beta)r_b}{48} > 0$, we drive $c_\Delta < \tilde{c_5}(\tilde{c_5} = \frac{-16+(13-19\beta)r_a+(19-13\beta)r_b}{6})$. Thus, if $c_\Delta < \tilde{c_5}$, $p_{an}^{BB*} > p_{ao}^{BB*}$. Otherwise, $p_{an}^{BB*} < p_{ao}^{BB*}$. In addition, let $p_{bn}^{BB*} - p_{bo}^{BB*} = \frac{-16+6c_\Delta+(19-13\beta)r_a+(13-19\beta)r_b}{48} > 0$. We can easily have: if $c_\Delta < \tilde{c_6}(\tilde{c_6} = \frac{16+(-19+13\beta)r_a+(-13+19\beta)r_b}{6})$, $p_{bn}^{BB*} > p_{bo}^{BB*}$. Otherwise, $p_{bn}^{BB*} < p_{bo}^{BB*}$.

**Proof of Proposition 1.** In order to compare the difference of results in Models NN and BN, let $\pi_{a2}^{BN*} - \pi_{a2}^{NN*} = \frac{-98+16c_\Delta^2+4(1+\beta)^2r_a^2+(7-2(1+\beta)r_b)^2+8c_\Delta(7+2(1+\beta)r_a-2(1+\beta)r_b)+4r_a(7(1-7c+\beta)-2(1+\beta)^2r_b)}{196} > 0$, we derive $c < \hat{c_1}(\hat{c_1} = \frac{-2(1+\beta)r_a-7(1+\sqrt{2+4cr_a})+2(1+\beta)r_b}{4})$. So, if $c < \hat{c_1}$, $\pi_{a2}^{BN*} > \pi_{a2}^{NN*}$. Otherwise, $\pi_{a2}^{BN*} < \pi_{a2}^{NN*}$.

Let $\pi_{b2}^{BN*} - \pi_{b2}^{NN*} = \frac{-98+(-7+4c_\Delta+2(1+\beta)r_a)^2-4(-7+49c-7\beta+4(1+\beta)c_\Delta+2(1+\beta)^2r_a)r_b+4(1+\beta)^2r_b^2}{196} > 0$, we derive $c < \bar{c_1}(\bar{c_1} = \frac{7-2(1+\beta)r_a+2(1+\beta)r_b+7\sqrt{2+4cr_b}}{4})$. Therefore, if $c < \bar{c_1}$, $\pi_{b2}^{BN*} > \pi_{b2}^{NN*}$. Otherwise, $\pi_{b2}^{BN*} < \pi_{b2}^{NN*}$. Let $p_{a2}^{BN*} - p_{a2}^{NN*} = \frac{-7+4c_\Delta+2(1+\beta)r_a-2(1+\beta)r_b}{14} < 0$ and $p_{b2}^{BN*} - p_{b2}^{NN*} = \frac{-7-4c_\Delta-2(1+\beta)r_a+2(1+\beta)r_b}{14} < 0$. As a result, $p_{a2}^{BN*} < p_{a2}^{NN*}$ $(p_{b2}^{BN*} < p_{b2}^{NN*})$ always holds.

**Proof of Proposition 2.** Accordingly, we compare the difference between them in period 1 and get $\pi_{a1}^{BN*} - \pi_{a1}^{NN*} = \frac{2r_a(-18(\beta+1)^2r_b+77\beta-294c+77)+4c_\Delta(18(\beta+1)r_a-18(\beta+1)r_b+77)+18(\beta+1)^2r_a^2+2(\beta+1)r_b(9(\beta+1)r_b-77)+72c_\Delta^2+49}{588}$. Let $\pi_{a1}^{BN*} > \pi_{a1}^{NN*}$, we derive $c_\Delta > \hat{c_3}$ $(\hat{c_3} = \frac{1}{36}\left(7\left(\sqrt{216cr_a+103}-11\right)-18(\beta+1)r_a+18(\beta+1)r_b\right))$. Consequently, if $c_\Delta > \hat{c_3}$, $\pi_{a1}^{BN*} > \pi_{a1}^{NN*}$. Otherwise, $\pi_{a1}^{BN*} < \pi_{a1}^{NN*}$. Let $p_{a1}^{BN*} - p_{a1}^{NN*} = \frac{3(\beta+1)r_a-3(\beta+1)r_b+6c_\Delta-7}{21} > 0$, we drive $c_\Delta > \tilde{c_7}(\tilde{c_7} = \frac{(-3(\beta+1)r_a+3(\beta+1)r_b+7)}{6})$. Thus, if $c_\Delta > c_\Delta > \tilde{c_7}$, $p_{a1}^{BN*} > p_{a1}^{NN*}$. Otherwise, $p_{a1}^{BN*} < p_{a1}^{NN*}$.

**Proof of Proposition 3.** In addition, we compare the equilibrium results in Models BN and BB. $\pi_{a2}^{BB*} - \pi_{a2}^{BN*} = \frac{12c_\Delta((767\beta-17)r_a+(17\beta-767)r_b+896)+2r_a((\beta(5147\beta-14794)+5147)r_b)+(\beta(9749\beta-10294)+5045)r_a^2+(\beta(5045\beta-10294)+9749)r_b^2+3584(2\beta-5)r_b+6272}{225792}$. Let $\pi_{a2}^{BB*} - \pi_{a2}^{BN*} > 0$, we drive $c_\Delta < \hat{c_4}$

$(\hat{c_4} = \frac{1}{750}\left(-767\beta r_a + 17r_a - 17\beta r_b + 767r_b - 896\right) - \frac{28\sqrt{-134\beta^2 r_a r_b + 268\beta r_a r_b - 134 r_a r_b - 67\beta^2 r_a^2 + 134\beta r_a^2 - 92\beta r_a - 67 r_a^2 + 92 r_a - 67\beta^2 r_b^2 + 134\beta r_b^2}}{125\sqrt{3}})$. $\pi_{b2}^{BB*} - \pi_{b2}^{BN*} = \frac{-12c_\Delta((17\beta-767)r_a+(767\beta-17)r_b+896)+2r_a((\beta(5147\beta-14794)+5147)r_b)+(\beta(5045\beta-10294)+9749)r_a^2+(\beta(9749\beta-10294)+5045)r_b^2+3584(5\beta-2)r_b+6272}{225792}$. Let $\pi_{b2}^{BB*} - \pi_{b2}^{BN*} > 0$, we drive $c_\Delta < \hat{c_5}$ $(\hat{c_5} = \frac{1}{750}\left(17\beta r_a - 767 r_a + 767\beta r_b - 17 r_b + 896\right) - \frac{28\sqrt{-134\beta^2 r_a r_b + 268\beta r_a r_b - 134 r_a r_b - 67\beta^2 r_a^2 + 134\beta r_a^2 - 92\beta r_a - 67 r_a^2 + 92 r_a - 67\beta^2 r_b^2 + 134\beta r_b^2 - 92\beta r_b - 67 r_b^2 + 92 r_b + 2}}{125\sqrt{3}})$. Thus, if $c_\Delta < \hat{c_4}$, $\pi_{a2}^{BB*} > \pi_{a2}^{BN*}$. Otherwise, $\pi_{a2}^{BB*} < \pi_{a2}^{BN*}$. If $c_\Delta < \hat{c_5}$, $\pi_{b2}^{BB*} > \pi_{b2}^{BN*}$. Otherwise, $\pi_{b2}^{BB*} < \pi_{b2}^{BN*}$.

From the perspective of price dimension, we obtain the following results.

Let $p_{ao}^{BB*} - p_{a2}^{BN*} = \frac{1}{336}\left((71\beta - 41)r_a + (41\beta - 71)r_b + 30c_\Delta + 56\right) > 0$, we have $c_\Delta > \bar{c_3}(\bar{c_3} = \frac{1}{30}\left(-71\beta r_a + 41 r_a - 41\beta r_b + 71 r_b - 56\right))$. If $c_\Delta > \bar{c_3}$, $p_{ao}^{BB*} > p_{a2}^{BN*}$. Otherwise, $p_{ao}^{BB*} < p_{a2}^{BN*}$.

Also, let $p_{an}^{BB*} - p_{a2}^{BN*} = \frac{1}{168}\left((25 - 31\beta)r_a + (31 - 25\beta)r_b - 6c_\Delta - 28\right) > 0$, we drive $c_\Delta < \bar{c_4}(\bar{c_4} = \frac{1}{6}\left(-31\beta r_a + 25 r_a - 25\beta r_b + 31 r_b - 28\right))$. Thus, if $c_\Delta < \bar{c_4}$, $p_{an}^{BB*} > p_{a2}^{BN*}$. Otherwise, $p_{an}^{BB*} < p_{a2}^{BN*}$.

Similarly, $p_{bo}^{BB*} - p_{a2}^{BN*} = \frac{1}{336}\left((41\beta - 71)r_a + (71\beta - 41)r_b - 30c_\Delta + 56\right) > 0$, we drive $c_\Delta < \bar{c_5}(\bar{c_5} = \frac{1}{30}\left(41\beta r_a - 71 r_a + 71\beta r_b - 41 r_b + 56\right))$. If $c_\Delta < \bar{c_5}$, $p_{bo}^{BB*} > p_{a2}^{BN*}$. Otherwise, $p_{bo}^{BB*} < p_{a2}^{BN*}$.

Finally, let $p_{bn}^{BB*} - p_{a2}^{BN*} = \frac{1}{168}\left((31 - 25\beta)r_a + (25 - 31\beta)r_b + 6c_\Delta - 28\right) > 0$, we drive $c_\Delta > \bar{c_6}(\bar{c_6} = \frac{1}{6}\left(-31 r_a - 25 r_b + 28\right))$. Thus, if $c_\Delta > \bar{c_6}$, $p_{bn}^{BB*} > p_{a2}^{BN*}$. Otherwise, $p_{bn}^{BB*} < p_{a2}^{BN*}$.

**Proof of Proposition 4.** Based on Model BN and Model BB, we can get the following results. Let $\pi_{a1}^{BB*} - \pi_{a1}^{BN*} = \frac{-36c_\Delta(33(\beta+1)r_a-112)+2r_a((\beta(6569\beta-11950)+6569)r_b+13552)+r_a^2+r_b((\beta(5975\beta-13138)+5975)r_b-27104\beta+23072)-1188c_\Delta^2+81536}{225792} > 0$. $\pi_{b1}^{BB*} - \pi_{b1}^{BN*} = \frac{2r_a((\beta(6569\beta-11950)+6569)r_b-13552\beta+11536)-36c_\Delta(33(\beta+1)r_a-33(\beta+1)r_b+112)+(\beta(5975\beta-13138)+5975)r_a^2+r_b((\beta(5975\beta-13138)+5975)r_b-23072\beta+27104)-1188c_\Delta^2}{225792} > 0$. We find, $\pi_{a1}^{BB*} > \pi_{a1}^{BN*}$ and $\pi_{b1}^{BB*} > \pi_{b1}^{BN*}$ always hold.

In the same way, let $p_{a1}^{BB*} - p_{a1}^{BN*} = \frac{1}{168}\left((53 - 59\beta)r_a + (59 - 53\beta)r_b - 6c_\Delta + 112\right) > 0$ and $p_{b1}^{BB*} - p_{b1}^{BN*} = \frac{1}{168}\left((59 - 53\beta)r_a + (53 - 59\beta)r_b + 6c_\Delta + 112\right) > 0$. We have, $p_{a1}^{BB*} > p_{a1}^{BN*}$ and $p_{b1}^{BB*} > p_{b1}^{BN*}$ always exist.

**Proof of Proposition 5.** We compare the equilibrium prices and profits in Models NN and BB. Let $\pi_{a2}^{BB*} - \pi_{a2}^{NN*} = \frac{12c_\Delta((47\beta+31)r_a-(31\beta+47)r_b+128)+2r_a((\beta(11\beta-490)+11)r_b-256(-2\beta))+(\beta(293\beta-22)+197)r_a^2+(\beta(197\beta-22)+293)r_b^2-512(\beta+2)r_b+468c_\Delta^2-1024}{4608} > 0$, we drive $c_\Delta > \bar{c_4}(\bar{c_4} = \frac{-8\left(\sqrt{4r_a(-5\beta+234c+5)-25r_a(\beta-1)^2(2r_b-r_a)}+16\right)-(47\beta+31)r_a}{78})$. So that, if $c_\Delta > \bar{c_4}$, $\pi_{a2}^{BB*} > \pi_{a2}^{NN*}$. Otherwise, $\pi_{a2}^{BB*} < \pi_{a2}^{NN*}$. $\pi_{b2}^{BB*} - \pi_{b2}^{NN*} = \frac{12c_\Delta((31\beta+47)r_a-(47\beta+31)r_b-128)-512((\beta+2)r_a+(9c-1)r_b+2)+2r_b((\beta(11\beta-490)+11)r_a+512\beta)+(\beta(197\beta-22)+293)r_a^2+(\beta(293\beta-22)+197)r_b^2+468c_\Delta^2}{4608} > 0$, we drive $c_\Delta > \hat{c_6}$. Thus, if $c_\Delta > \hat{c_6}$, $\pi_{b2}^{BB*} > \pi_{b2}^{NN*}$. Otherwise, $\pi_{b2}^{BB*} < \pi_{b2}^{NN*}$. $(\hat{c_6} = \frac{1}{78}\left(-8\sqrt{-10(\beta-1)r_a\left(5(\beta-1)r_b\right) - 25(\beta-1)^2 r_a^2 + 4r_b(-5\beta) + 464} - (31\beta + 47)r_a + r_b + 128\right))$.

Also, let $p_{ao}^{BB*} - p_{a2}^{NN*} = \frac{1}{48}\left((17\beta + 1)r_a - (\beta + 17)r_b + 18c_\Delta - 16\right) > 0$, we have $c_\Delta > \tilde{c_9}(\tilde{c_9} = \frac{1}{18}\left(-(17\beta + 1)r_a + (\beta + 17)r_b + 16\right))$. Therefore, if $c_\Delta > \tilde{c_9}$, $p_{ao}^{BB*} > p_{a2}^{NN*}$. Otherwise, $p_{ao}^{BB*} < p_{a2}^{NN*}$. Let $p_{an}^{BB*} - p_{a2}^{NN*} = \frac{1}{24}\left(-(\beta - 7)r_a - 7\beta r_b + r_b + 6c_\Delta - 16\right) > 0$, $p_{an}^{BB*} > p_{a2}^{NN*}$ always holds. Similarly, let $p_{bo}^{BB*} - p_{a2}^{NN*} = \frac{1}{48}\left(-(\beta + 17)r_a + 17\beta r_b + r_b - 18c_\Delta - 16\right) > 0$, we drive $c_\Delta < \tilde{c_{10}}(\tilde{c_{10}} = \frac{1}{18}\left(-(\beta + 17)r_a + (17\beta + 1)r_b - 16\right))$. Thus, if $c_\Delta < \tilde{c_{10}}$, $p_{bo}^{BB*} > p_{a2}^{NN*}$. Otherwise, $p_{bo}^{BB*} < p_{a2}^{NN*}$. Meanwhile, let $p_{bn}^{BB*} - p_{a2}^{NN*} = \frac{1}{24}\left((1 - 7\beta)r_a - (\beta - 7)r_b - 6c_\Delta - 16\right) > 0$. So, $p_{bn}^{BB*} < p_{a2}^{NN*}$ always holds.

**Proof of Proposition 6.** Based on Model NN and Model BB, their comparison is as follows. Let $\pi_{a1}^{BB*} - \pi_{a1}^{NN*} = \frac{12c_\Delta(45(\beta+1)r_a-45(\beta+1)r_b+208)+2r_a(-(\beta(7\beta+526)+7)r_b+368\beta+880)+(\beta(263\beta+14)+263)r_a^2+(\beta(263\beta+14)+263)r_b^2-32(55\beta+23)r_b+540c_\Delta^2+2048}{4608} > 0$, we drive $c_\Delta > \bar{c_5}$. So, if $c_\Delta > \bar{c_5}$, $\pi_{a1}^{BB*} > \pi_{a1}^{NN*}$. Otherwise, $\pi_{a1}^{BB*} < \pi_{a1}^{NN*}$. Meanwhile, we have $\pi_{b1}^{BB*} - \pi_{b1}^{NN*} = \frac{12c_\Delta(45(\beta+1)r_a-45(\beta+1)r_b-208)-2r_a((\beta(7\beta+526)+7)r_b+880\beta+368)+(\beta(263\beta+14)+263)r_a^2+32r_b(23\beta-144c+55)+(\beta(263\beta+14)+263)r_b^2+540c_\Delta^2+2048}{4608} > 0$, we drive $c_\Delta < \bar{c_6}$. If $c_\Delta < \bar{c_6}$, $\pi_{b1}^{BB*} > \pi_{b1}^{NN*}$. Otherwise, $\pi_{b1}^{BB*} < \pi_{b1}^{NN*}$. $(\bar{c_5} = \frac{(-45\beta r_a-45r_a+45\beta r_b+45r_b+208)-\frac{4}{45}\sqrt{2}\sqrt{-30\beta^2 r_a r_b+60\beta r_a r_b-30 r_a r_b-15\beta^2 r_a^2+30\beta r_a^2}}{90})$.
$(\bar{c_6} = \frac{\left(-8\sqrt{2}\sqrt{-30(\beta-1)r_a((\beta-1)r_b-2)-15(\beta-1)^2 r_a^2-15(\beta-1)^2 r_b^2+98+45(\beta+1)r_b+208}\right)}{90})$.

From the perspective of price dimension, we obtain the following results. Let $p_{a1}^{BB*} - p_{a1}^{NN*} = \frac{1}{24}\left((11 - 5\beta)r_a + (5 - 11\beta)r_b + 6c_\Delta + 8\right) > 0$, we drive $c_\Delta > \hat{c_7}(\hat{c_7} = \frac{1}{6}\left(5\beta r_a - 11 r_a + 11\beta r_b - 5 r_b - 8\right))$. Thus, if $c_\Delta > \hat{c_7}$, $p_{a1}^{BB*} > p_{a1}^{NN*}$. Otherwise, $p_{a1}^{BB*} < p_{a1}^{NN*}$. In the same way, let $p_{b1}^{BB*} - p_{b1}^{NN*} = \frac{1}{24}\left((5 - 11\beta)r_a + (11 - 5\beta)r_b - 6c_\Delta + 8\right) > 0$, we drive $c_\Delta > \hat{c_8}(\hat{c_8} = \frac{1}{6}\left((5 - 11\beta)r_a + (11 - 5\beta)r_b + 8\right))$. So, if $c_\Delta > \hat{c_8}$, $p_{b1}^{BB*} > p_{b1}^{NN*}$. Otherwise, $p_{b1}^{BB*} < p_{b1}^{NN*}$.

**Proof of Corollary 3.** We take the first-order derivatives of $\pi_{a2}^{BN}$, with respect to $r_a$, $\frac{\partial \pi_{a2}^{BN*}}{\partial r_a} = \frac{1}{49}\left(2(\beta + 1)^2 r_a - 2(\beta + 1)^2 r_b + 4(\beta + 1)c_\Delta + 7(\beta - 7c + 1)\right)$. Let $\frac{\partial \pi_{a2}^{BN*}}{\partial r_a} > 0$, we drive $r_a > \tilde{r_1}$ $(\tilde{r_1} = r_b + \frac{49c-(\beta+1)(4c_\Delta+7)}{2(\beta+1)^2})$, otherwise, $\frac{\partial \pi_{a2}^{BN*}}{\partial r_a} < 0$.

In addition, let $\frac{\partial \pi_{b2}^{BN*}}{\partial r_b} = \frac{1}{49}\left(-2(\beta + 1)^2 r_a + 2(\beta + 1)^2 r_b - 4(\beta + 1)c_\Delta + 7(\beta - 7c + 1)\right) > 0$, we derive $r_b > \tilde{r_2}$ $(\tilde{r_2} = r_a + \frac{49c+(\beta+1)(4c_\Delta-7)}{2(\beta+1)^2})$, otherwise, $\frac{\partial \pi_{b2}^{BN*}}{\partial r_a} < 0$.

Similarly, we take the first-order derivatives of $p_{a2}^{BN}$, with respect to $r_a$, let $\frac{\partial p_{a2}^{BN*}}{\partial r_a} = \frac{1+\beta}{7} > 0$ and $\frac{\partial p_{b2}^{BN*}}{\partial r_b} = \frac{1+\beta}{7} > 0$. Since $0 < \beta < 1$, $\frac{\partial p_{a2}^{BN*}}{\partial r_a} > 0$ and $\frac{\partial p_{b2}^{BN*}}{\partial r_b} > 0$ always hold.

**Proof of Corollary 4.** Taking the first-order derivatives, we have the following results. Let $\frac{\partial \pi_{a1}^{BN*}}{\partial r_a} = \frac{1}{294}\left(18(\beta + 1)^2 r_a - 18(\beta + 1)^2 r_b + 77\beta + 36(\beta + 1)c_\Delta - 294c + 77\right) > 0$, we drive $r_a > \tilde{r_3}$ $(\tilde{r_3} = r_b + \frac{294c-(\beta+1)(36c_\Delta+77)}{18(\beta+1)^2})$, otherwise, $\frac{\partial \pi_{a1}^{BN*}}{\partial r_a} < 0$.

Also, let $\frac{\partial \pi_{b1}^{BN*}}{\partial r_b} = \frac{1}{294}\left(-18(\beta+1)^2 r_a + 18(\beta+1)^2 r_b + 77\beta - 36(\beta+1)c_\Delta - 294c + 77\right) > 0$, we drive $r_b > \tilde{r_4}$ ($\tilde{r_4} = r_a + \frac{294c+(\beta+1)(36c_\Delta-77)}{18(\beta+1)^2}$), otherwise, $\frac{\partial \pi_{b1}^{BN*}}{\partial r_b} < 0$.

Similarly, $\frac{\partial p_{a2}^{BN*}}{\partial r_a} = \frac{1+\beta}{7} > 0$ and $\frac{\partial p_{b2}^{BN*}}{\partial r_b} = \frac{1+\beta}{7} > 0$ always hold for $0 < \beta < 1$.

**Proof of Corollary 5.** Like the proof of Corollary 4, we take the first-order derivatives and let $\frac{\partial \pi_{a2}^{BB*}}{\partial r_a} = \frac{(\beta(293\beta-22)+197)r_a+(\beta(11\beta-490)+11)r_b+6(47\beta+31)c_\Delta-256(-2\beta+9c-1)}{2304} > 0$, we drive $r_a > \tilde{r_5}$ ($\tilde{r_5} = \frac{((490-11\beta)\beta-11)r_b+2304c-2(256\beta+141\beta c_\Delta+93c_\Delta+128)}{\beta(293\beta-22)+197}$), otherwise, $\frac{\partial \pi_{a2}^{BB*}}{\partial r_a} < 0$.

Similarly, $\frac{\partial \pi_{b2}^{BB*}}{\partial r_b} = \frac{(\beta(11\beta-490)+11)r_a+(\beta(293\beta-22)+197)r_b-6(47\beta+31)c_\Delta-256(-2\beta+9c-1)}{2304} > 0$, we derive $r_b > \tilde{r_6}$ ($\tilde{r_6} = \frac{((490-11\beta)\beta-11)r_a-256(2\beta+1)+2304c+6(47\beta+31)c_\Delta}{\beta(293\beta-22)+197}$), otherwise, $\frac{\partial \pi_{b2}^{BB*}}{\partial r_b} < 0$.

Also, we take first-order derivatives to the prices and get the results. $\frac{\partial p_{ao}^{BB*}}{\partial r_a} = \frac{1}{48}(17\beta+1)$, $\frac{\partial p_{an}^{BB*}}{\partial r_a} = \frac{7-\beta}{24}$, $\frac{\partial p_{bo}^{BB*}}{\partial r_b} = \frac{1}{48}(17\beta+1)$, and $\frac{\partial p_{bn}^{BB*}}{\partial r_b} = \frac{7-\beta}{24}$. Therefore, $\frac{\partial p_{ao}^{BB*}}{\partial r_a} > 0$, $\frac{\partial p_{an}^{BB*}}{\partial r_a} > 0$, $\frac{\partial p_{bo}^{BB*}}{\partial r_b} > 0$ and $\frac{\partial p_{bn}^{BB*}}{\partial r_b} > 0$ always hold.

**Proof of Corollary 6.** Based on the equilibrium results of Model BN, we analyze the influence of the privacy cost difference. Let $\frac{\partial \pi_{a2}^{BN*}}{\partial c_\Delta} = \frac{2}{49}\left(2(\beta+1)r_a - 2(\beta+1)r_b + 4c_\Delta + 7\right) > 0$, we drive $c_\Delta > c_{\Delta 1}^{\wedge}$ ($c_{\Delta 1}^{\wedge} = \frac{1}{4}\left(-2(\beta+1)r_a + 2(\beta+1)r_b - 7\right)$), otherwise, $\frac{\partial \pi_{a2}^{BN*}}{\partial c_\Delta} < 0$.

In addition, let $\frac{\partial \pi_{b2}^{BN*}}{\partial c_\Delta} = \frac{2}{49}\left(2(\beta+1)r_a - 2(\beta+1)r_b + 4c_\Delta + 7\right) > 0$, we derive $c_\Delta > c_{\Delta 2}^{\wedge}$ ($c_{\Delta 2}^{\wedge} = \frac{1}{4}\left(-2(\beta+1)r_a + 2(\beta+1)r_b - 7\right)$), otherwise, $\frac{\partial \pi_{b2}^{BN*}}{\partial c_\Delta} < 0$. Similarly, $\frac{\partial p_{a2}^{BN*}}{\partial c_\Delta} = \frac{2}{7}$. $\frac{\partial p_{b2}^{BN*}}{\partial c_\Delta} = -\frac{2}{7}$. It can be seen that $\frac{\partial p_{a2}^{BN*}}{\partial c_\Delta} > 0$ and $\frac{\partial p_{b2}^{BN*}}{\partial c_\Delta} < 0$ always hold.

**Proof of Corollary 7.** Taking the first-order derivatives, we find

$\frac{\partial \pi_{a1}^{BN*}}{\partial c_\Delta} = \frac{1}{147}\left(18(\beta+1)r_a - 18(\beta+1)r_b + 36c_\Delta + 77\right) > 0$,

$\frac{\partial \pi_{b1}^{BN*}}{\partial c_\Delta} = \frac{1}{147}\left(18(\beta+1)r_a - 18(\beta+1)r_b + 36c_\Delta - 77\right) < 0$.

Thus, $\frac{\partial \pi_{a1}^{BN*}}{\partial c_\Delta} > 0$ and $\frac{\partial \pi_{b1}^{BN*}}{\partial c_\Delta} < 0$ always hold. Furthermore, $\frac{\partial p_{a1}^{BN*}}{\partial c_\Delta} = \frac{2}{7}$, $\frac{\partial p_{b1}^{BN*}}{\partial c_\Delta} = -\frac{2}{7}$. So that, $\frac{\partial p_{a1}^{BN*}}{\partial c_\Delta} > 0$ and $\frac{\partial p_{b1}^{BN*}}{\partial c_\Delta} < 0$ always hold.

**Proof of Corollary 8.** Finally, we analyze the influence of $c_\Delta$ in Model BB.

For retailer A, let $\frac{\partial \pi_{a2}^{BB*}}{\partial c_\Delta} = \frac{1}{384}\left((47\beta+31)r_a - (31\beta+47)r_b + 78c_\Delta + 128\right) > 0$, we drive $c_\Delta > \hat{c}_{\Delta 3}$ ($\hat{c}_{\Delta 3} = \frac{1}{78}\left(-(47\beta+31)r_a + (31\beta+47)r_b - 128\right)$), otherwise, $\frac{\partial \pi_{a2}^{BB*}}{\partial c_\Delta} < 0$.

For retailer B, let $\frac{\partial \pi_{b2}^{BB*}}{\partial c_\Delta} = \frac{1}{384}\left((31\beta+47)r_a - (47\beta+31)r_b + 78c_\Delta - 128\right) > 0$, we derive $c_\Delta > \hat{c}_{\Delta 4}$ ($\hat{c}_{\Delta 4} = \frac{1}{78}\left(-(31\beta+47)r_a + (47\beta+31)r_b + 128\right)$), otherwise, $\frac{\partial \pi_{b2}^{BB*}}{\partial c_\Delta} < 0$.

Also, we get $\frac{\partial p_{ao}^{BB*}}{\partial c_\Delta} = \frac{3}{8}$, $\frac{\partial p_{an}^{BB*}}{\partial c_\Delta} = \frac{1}{4}$, $\frac{\partial p_{bo}^{BB*}}{\partial c_\Delta} = -\frac{3}{8}$ and $\frac{\partial p_{bn}^{BB*}}{\partial c_\Delta} = -\frac{1}{4}$. Thus, $\frac{\partial p_{ao}^{BB*}}{\partial c_\Delta} > 0$, $\frac{\partial p_{an}^{BB*}}{\partial c_\Delta} > 0$, $\frac{\partial p_{bo}^{BB*}}{\partial c_\Delta} < 0$ and $\frac{\partial p_{bn}^{BB*}}{\partial c_\Delta} < 0$ always hold.

**Proof of Proposition 7.** We compare the prices of three models and obtain the following results. According to the previous proof, we have $p^{NN} = 1$, $p^{BN} = \frac{7+4c_\Delta+2(1+\beta)r_a-2(1+\beta)r_b}{14}$ and $p^{BB} = \frac{5(\beta+1)r_a-5(\beta+1)r_b+10c_\Delta+16}{16}$. If $c_\Delta < \hat{c}_{\Delta 5}$, $p^{NN} > p^{BB} > p^{BN}$, otherwise, $p^{BB} > p^{NN} > p^{BN}$. ($\hat{c}_{\Delta 5} = \frac{1}{18}\left(-(17\beta+1)r_a + (\beta+17)r_b + 16\right)$). We let $r_a = 0.3, r_b = 0.5, \beta = 0.5$ substitute into the above equations and then draw Fig. 4(a) after comparing them.

Similar to the previous proposition, based on the analytic solution in the previous section. We have $\pi^{NN} = \frac{1}{2}$, $\pi^{BN} = \frac{\left(4r_a\left(7(\beta-7c_\Delta+1)-2(\beta+1)^2 r_b\right)+8c_\Delta(2(\beta+1)r_a-2(\beta+1)r_b+7)+4(\beta+1)^2 r_a^2+(7-2(\beta+1)r_b)^2+16c_\Delta^2\right)}{196}$ and $\pi^{BB} = \frac{512r_a(2\beta-9c+1)+12c_\Delta((47\beta+31)r_a-31\beta r_b-47r_b+128)+(\beta(293\beta-22)+197)r_a^2+2(\beta(11\beta-490)+11)r_a r_b+(\beta(197\beta-22)+293)r_b^2-512(\beta+2)r_b+468c_\Delta^2+1280}{4608}$. By subtracting the profits, we find some outcomes. If $c_\Delta < \hat{c}_{\Delta 7}, \pi^{NN} > \pi^{BN} > \pi^{BB}$. If $\hat{c}_{\Delta 6} > c_\Delta > \hat{c}_{\Delta 7}, \pi^{NN} > \pi^{BB} > \pi^{BN}$. If $\hat{c}_{\Delta 8} > c_\Delta > \hat{c}_{\Delta 6}, \pi^{BB} > \pi^{NN} > \pi^{BN}$. If $c_\Delta > \hat{c}_{\Delta 8}, \pi^{BB} > \pi^{BN} > \pi^{NN}$. Thus, if $c_\Delta < \hat{c}_{\Delta 6}, \pi^{NN}$ is higher than $\pi^{BB}$ and $\pi^{BN}$. Otherwise, $\pi^{BB}$ is higher than $\pi^{BN}$ and $\pi^{NN}$. $\hat{c}_{\Delta 6} = \frac{\left(-8\left(\sqrt{\alpha-25(\beta-1)^2 r_a^2-5(\beta-1)r_b(5(\beta-1)r_b+4)+464}\right)-(47\beta+31)r_a+(31\beta+47)r_b\right)}{78}$ ($\alpha = 2r_a\left(2(-5\beta+234c+5)-25(\beta-1)^2 r_b\right)$), $\hat{c}_{\Delta 7} = \frac{1}{750}\left(56\left(\sqrt{\eta}\right)+(17-767\beta)r_a+(767-17\beta)r_b\right)$ ($\eta = -6(\beta-1)r_a\left(67(\beta-1)r_b+46\right)-201(\beta-1)^2 r_a^2-3(\beta-1)r_b\left(67(\beta-1)r_b+92\right)+6-16$) and $\hat{c}_{\Delta 8} = \frac{1}{4}\left(-7\left(\sqrt{4cr_a+2}+1\right)-2(\beta+1)r_a+2(\beta+1)r_b\right)$. We substitute $r_a = 0.3, r_b = 0.5, \beta = 0.5$ into the following equation, draw Fig. 4(b).

**Proof of Extension 7.1.** With backward induction, we obtained the corresponding analytical solution through computation. Retailer A set prices at different levels in the second and first periods respectively: $p_{a2}^{BN*} = \frac{2(\beta+1)\lambda(r_b-r_a)-4\lambda c_\Delta+2\lambda-9}{(\lambda+1)(2\lambda-9)}$ and $p_{a1}^{BN*} = \frac{2\lambda(-3(\beta+1)r_a+3(\beta+1)r_b-6c_\Delta+4\lambda-18)}{3(\lambda+1)(2\lambda-9)}$. Retailer B's pricing in the second and first periods are, respectively: $p_{b2}^{BN*} = \frac{2(\beta+1)\lambda(r_a-r_b)+4\lambda c_\Delta+2\lambda-9}{(\lambda+1)(2\lambda-9)}$ and $p_{b1}^{BN*} = \frac{2\lambda(3(\beta+1)r_a-3(\beta+1)r_b+6c_\Delta+4\lambda-18)}{3(\lambda+1)(2\lambda-9)}$.

By employing backward induction, optimal pricing strategies for retailers A and B in the second period can be computed. The same methodology is applied to determine pricing strategies in the first period, accounting for consumer price expectations and responses to personalized recommendations. Retailers A and B set prices in the second and first periods as follows:

$p_{ao}^{BB*} = \frac{\lambda r_a(\beta(\lambda(\lambda(5\lambda-51)+81)-18)+\lambda((\lambda-3)\lambda-3)+6)+\lambda(-(\beta+5)\lambda^3+3(\beta+17)\lambda^2+3(\beta-27)\lambda-6(\beta-3))r_b+6\lambda(\lambda((\lambda-9)\lambda+13)-2)c_\Delta}{12\lambda(\lambda((\lambda-12)\lambda+21)-6)}$,

$p_{an}^{BB*} = \frac{1}{6}\left(\frac{r_a(-\beta(\lambda((\lambda-21)\lambda+27)-6)+\lambda((\lambda-3)\lambda+15)-6)+(-\beta(\lambda((\lambda-3)\lambda+15)-6)+\lambda((\lambda-21)\lambda+27)-6)r_b+6\lambda(3\lambda-2)c_\Delta}{\lambda((\lambda-12)\lambda+21)-6}\right)$,

$p_{bo}^{BB*} = \frac{\lambda r_a(-(\beta+5)\lambda^3+3(\beta+17)\lambda^2+3(\beta-27)\lambda-6(\beta-3))+\lambda(\beta(\lambda(\lambda(5\lambda-51)+81)-18)+\lambda(\lambda-3)+6)r_b-6\lambda(\lambda((\lambda-9)\lambda+13))c_\Delta}{12\lambda(\lambda((\lambda-12)\lambda+21)-6)}$,

$$p_{bn}^{BB*} = \frac{1}{6}\left(\frac{r_a(-\beta(\lambda((\lambda-3)\lambda+15)-6)+\lambda((\lambda-21)\lambda+27)-6)+(-\beta(\lambda((\lambda-21)\lambda+27)-6)+\lambda((\lambda-3)\lambda+15)-6)r_b+6\lambda(2-3\lambda)c_\Delta}{\lambda((\lambda-12)\lambda+21)-6}\right),$$
$$p_{a1}^{BB*} = \frac{\lambda r_a(\lambda(-\beta(\lambda-6)(\lambda((\lambda-15)\lambda+23)-6)+156)-24)+r_b+6\lambda(3\lambda-2)(\lambda(2\lambda-7)+2)c_\Delta-2(\lambda((\lambda-12)\lambda+21)-6)(\lambda(\lambda-17)+2)}{6(\lambda(2\lambda-7)+2)(\lambda((\lambda-12)\lambda+21)-6)},$$
$$p_{b1}^{BB*} = \frac{\lambda r_a((\lambda-6)\lambda(\lambda((\lambda-15)\lambda+23)-6))+\lambda r_b-6\lambda(3\lambda-2)(\lambda(2\lambda-7)+2)c_\Delta-2(\lambda((\lambda-12)\lambda+21)-6)(\lambda(\lambda(2(\lambda-8)\lambda+41)-17)+2)}{6(\lambda(2\lambda-7)+2)(\lambda((\lambda-12)\lambda+21)-6)}.$$

**Proof of Extension 7.2.** By computation, we derive the optimal pricing strategies for retailers A and B under conditions where personalized recommendations and BBP are not feasible. Then, $p_{\mathrm{a1}}^{DN*} = \frac{1}{3}\left(r_a - r_b + c_\Delta + 3\right)$, $p_{\mathrm{b1}}^{DN*} = \frac{1}{3}\left(-r_a + r_b - c_\Delta + 3\right)$, $p_{\mathrm{a2}}^{DN*} = \frac{1}{18}\left((3\beta+2)r_a - (3\beta+2)r_b + 5c_\Delta + 9\right)$, $p_{\mathrm{b2}}^{DN*} = \frac{1}{18}\left(-(3\beta+2)r_a + (3\beta+2)r_b - 5c_\Delta + 9\right)$. These pricing strategies are equilibrium-based on market demand, costs, and consumer preferences.

We derive the optimal pricing strategies for retailers A and B by computation. Therefore, we have $p_{\mathrm{a1}}^{ND*} = p_{\mathrm{b1}}^{ND*} = 1$, $p_{\mathrm{a2}}^{ND*} = [(\beta+1)r_a - (\beta+1)r_b + 2c_\Delta + 3]/6$ and $p_{\mathrm{b2}}^{ND*} = [-(\beta+1)r_a + (\beta+1)r_b - 2c_\Delta + 3]/6$. Lastly, we analyze the scenario where retailers refrained from recommending in the first period, while concurrently engaging in personalized recommendations and BBP in the second period. Similarly, we get $p_{\mathrm{a1}}^{DD*} = p_{\mathrm{b1}}^{DD*} = 1$, $p_{\mathrm{ao}}^{DD*} = \frac{1}{3}\left(\beta r_a - r_b + c_\Delta + 2\right)$ and $p_{\mathrm{an}}^{DD*} = \frac{1}{3}\left(r_a - \beta r_b + c_\Delta + 1\right)$, $p_{\mathrm{bo}}^{DD*} = \frac{1}{3}\left(-r_a + \beta r_b - c_\Delta + 2\right)$ and $p_{\mathrm{bn}}^{DD*} = \frac{1}{3}\left(-\beta r_a + r_b - c_\Delta + 1\right)$.

## Data availability

No data was used for the research described in the article.

## References


[1] Montes R, Sand-Zantman W, Valletti T. The value of personal information in online markets with endogenous privacy. Manage Sci 2019;65(3):1342–62.
[2] Lin X, Zhou Y, Xie W, Zhong Y, Cao B. Pricing and product-bundling strategies for e-commerce platforms with competition. European J Oper Res 2020;283(3):1026–39.
[3] Wang J, Fan X, Zhang T. Behaviour-based pricing and wholesaling contracting under supply chain competition. J Oper Res Soc 2021;72(9):1943–59.
[4] Choe C, King S, Matsushima N. Pricing with cookies: Behavior-based price discrimination and spatial competition. Manage Sci 2018;64(12):5669–87.
[5] Chen C, Duan Y, Li G. Adoption of personalized pricing in a supply chain. Manag Decis Econ 2022;43(7):2715–28.
[6] Valletti T, Wu J. Consumer profiling with data requirements: Structure and policy implications. Prod Oper Manage 2020;29(2):309–29.
[7] Kadziński M, Wójcik M, Ciomek K. Review and experimental comparison of ranking and choice procedures for constructing a univocal recommendation in a preference disaggregation setting. Omega 2022;113:102715.
[8] Zhang C, Zhang N, Su W, Balezentis T. Online commodity recommendation model for interaction between user ratings and intensity-weighted hierarchical sentiment: A case study of LYCOM. Omega 2024;129:103161.
[9] Bandara R, Fernando M, Akter S. Privacy concerns in E-commerce: A taxonomy and a future research agenda. Electron Mark 2020;30(3):629–47.
[10] Ma P, Lu Y, Wang H, Wen D. Retailers' information disclosure strategies with behavior-based pricing in competitive supply chains. Manag Decis Econ 2023;44(5):2973–97.
[11] Zhang Y, Yang R, Yi Z, Yao S, Jing Y. Behavior-based pricing under asymmetric channel competition with strategic consumers. Omega 2023;120:102–916.
[12] Wang Y, Li M, Feng H, Feng N. Which is better for competing firms with quality increasing: Behavior-based price discrimination or uniform pricing? Omega 2023;118:102870.
[13] Li J, Ji X, Perera SC. Behaviour-based pricing for multi-version information goods. Int J Prod Res 2023;23:1–26.
[14] Yan B, Chen Z, Wang X, Jin Z. Influence of logistic service level on multichannel decision of a two-echelon supply chain. Int J Prod Res 2020;58(11):3304–29.
[15] Duan Y, Ge Y, Feng Y. Pricing and personal data collection strategies of online platforms in the face of privacy concerns. Electron Commerce Res 2022;22:539–59.
[16] Wu X, Liao H. Value-driven preference disaggregation analysis for uncertain preference information. Omega 2023;115:102793.
[17] Nie F, Li X, Zhou C. Impact of privacy regulation involving information collection on the ride-hailing market. Omega 2024;126:103077.
[18] Basu A, Bhaskaran S, Mukherjee R. Compatibility and information asymmetry in online matching platforms. Manage Sci 2024;Articles in Advance:1–20.
[19] Choi SH, Kang S, Jeon YJ. Personalized recommendation system based on product specification values. Expert Syst Appl 2006;31(3):607–16.
[20] Qian X, Feng H, Zhao G, Mei T. Personalized recommendation combining user interest and social circle. IEEE Trans Knowl Data Eng 2013;26(7):1763–77.
[21] Zhou W, Han W. Personalized recommendation via user preference matching. Inf Process Manage 2019;56(3):955–68.
[22] Cheah J, Lim X, Ting H, Liu Y, Quach S. Are privacy concerns still relevant? Revisiting consumer behaviour in omnichannel retailing. J Retail Consumer Serv 2022;65:102–242.
[23] Zhou C, Leng M, Liu Z, Cui X, Yu J. The impact of recommender systems and pricing strategies on brand competition and consumer search. Electron Commer Res Appl 2022;53:101–44.
[24] Alzaidi MS, Agag G. The role of trust and privacy concerns in using social media for e-retail services: The moderating role of COVID-19. J Retail Consumer Serv 2022;68. 103–042.
[25] Guo W, Tian J, Li M. Price-aware enhanced dynamic recommendation based on deep learning. J Retail Consumer Serv 2023;75:1–13.
[26] Hallikainen H, Luongo M, Dhir A, Laukkanen T. Consequences of personalized product recommendations and price promotions in online grocery shopping. J Retail Consumer Serv 2022;69:1–10.
[27] Villas-Boas JM. Dynamic competition with customer recognition. Rand J Econ 1999;30(4):604–31.
[28] Villas-Boas JM. Price cycles in markets with customer recognition. Rand J Econ 2004;35(3):486–501.
[29] Cui TH, Raju JS, Zhang ZJ. A price discrimination model of trade promotions. Mark Sci 2008;27(5):779–95.
[30] Zhang J. The perils of behavior-based personalization. Mark Sci 2011;30(1):170–86.
[31] Herweg F, Müller D. Price discrimination in input markets: Downstream entry and efficiency. J Econ Manag Strategy 2012;21(3):773–99.
[32] Li KJ, Jain S. Behavior-based pricing: An analysis of the impact of peer-induced fairness. Manage Sci 2016;62(9):2705–21.
[33] Li KJ. Behavior-based pricing in marketing channels. Mark Sci 2018;37(2):310–26.
[34] Klein G. Transformational and transactional leadership, organizational support and environmental competition intensity as antecedents of intrapreneurial behaviors. Eur Res Manag Bus Econ 2023;29(2):100–215.
[35] Zheng S, Zheng Q, Vakharia AJ. Wholesale price discrimination and contract unobservability. Prod Oper Manage 2024;33(6):1320–34.
[36] Veiga A. Price discrimination in selection markets. Rev Econ Stat 2023;Articles in Advance:1–45.
[37] Cheng M, Wu S, Zhang J, Yu X. Unlocking the advantages of differential pricing: A two-period model integrating consumer identifiable information and advertising effort. Manag Decis Econ 2023;45(2):1119–33.
[38] Zhou C, Xu Y, Ren Y, Yu J. Strategic adoption of the recommender system under online retailer competition and consumer search. Electron Commer Res Appl 2024;64:101–376.
[39] Acquisti A, Varian HR. Conditioning prices on purchase history. Mark Sci 2005;24(3):367–81.
[40] Goldfarb A, Tucker CE. Privacy regulation and online advertising. Manage Sci 2011;57(1):57–71.
[41] Conitzer V, Taylor CR, Wagman L. Hide and seek: Costly consumer privacy in a market with repeat purchases. Mark Sci 2012;31(2):277–92.
[42] Casadesus-Masanell R, Hervas-Drane A. Competing with privacy. Manage Sci 2015;61(1):229–46.
[43] Shy O, Stenbacka R. Customer privacy and competition. J Econ Manag Strat 2016;25(3):539–62.
[44] Ichihashi S. Online privacy and information disclosure by consumers. Amer Econ Rev 2020;110(2):569–95.
[45] Chen C, Duan Y. Impact of personalization and privacy concerns on information disclosure and pricing. J Retail Consumer Serv 2022;69. 103–099.
[46] Li KJ. Behavior-based quality discrimination. Manuf Service Oper Manag 2021;23(2):425–36.
[47] Jiang Y, Ji X, Wu J, Lu W. Behavior-based pricing and consumer fairness concerns with green product design. Ann Oper Res 2023;2:1–27.
[48] Li X, Li KJ, Wang X. Transparency of behavior-based pricing. J Mar Res 2020;57(1):78–99.
[49] Wu Z, Xu G, Chen Q, Mao N. Two stochastic optimization methods for shift design with uncertain demand. Omega 2023;115:102789.
[50] Du S, Sheng J, Peng J, Zhu Y. Competitive implications of personalized pricing with a dominant retailer. Transp Res E: Logist Transp Rev 2022;161:102–690.
[51] Garella PG, Laussel D, Resende J. Behavior based price personalization under vertical product differentiation. Int J Ind Organ 2021;76:102–717.
[52] Yao C, Mo Y. How does cross-platform externality impact pricing strategies? A two-stage discriminatory pricing model analysis. Manag Decis Econ 2024;45(3):1454–79.
[53] Li X, Li KJ. Beating the algorithm: Consumer manipulation, personalized pricing, and big data management. Manuf Serv Oper Manag 2023;25(1):36–49.